Racial Discrimination against Crime Victims: Evidence from Violent Crime Investigations

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Abstract

I develop and apply a new test for a consequential but overlooked form of racial discrimination: by police detectives against victims of crime. I test whether detectives display racial bias using a marginal outcome test derived from a model of police investigations. By performing the test on novel data on investigations of violent crime conducted by the Chicago Police Department over the last two decades, I find evidence of racial bias against Black victims: cases are 5.8 pp (6.4%) more likely to be rejected by the prosecution for insufficient evidence if the victim is Black rather than White. This is true for both fatal and non-fatal violent crimes, and holds true regardless the race of the alleged perpetrator. This disparity persists when focusing on court: violent crimes committed against Black victims are 6.7 pp (8.8%) less likely to result in a guilty ruling. I investigate the mechanisms behind the main results by leveraging the characteristics of the detective in charge at the time of case closure. Using several proxies for on-the-job experience, I find that although detectives appear to learn over time, this learning does not differ by victim race and therefore does not close the acceptance gaps, suggesting that the disparities are more likely driven by biased preferences, i.e. racial aniums. These differences are therefore consistent with a greater tolerance for failures to deliver justice when the victim is Black, i.e. a lower cost of type I error. These results suggest that efforts to increase oversight and accountability in policing should also extend to police investigations and the broader differential treatment of crime victims.

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1 Introduction

The overly punitive approach of US police towards minority civilians suspected of a crime has been a major topic of discussion in recent years (CNN, 2014; National Academies of Sciences and Medicine, 2022). Research has shown that non-White civilians are more likely to be stopped and searched for contraband or weapons (Antonovics and Knight, 2009; Anwar and Fang, 2006), receive a traffic ticket (Goncalves and Mello, 2021), be arrested (Raphael and Rozo, 2019), and subject to the use of force (Hoekstra and Sloan, 2022; Lieberman, 2024) by the police.[1] Related work has documented that these forms of over-sanctioning by the police significantly erode trust in law enforcement institutions (Desmond et al., 2016; Ang et al., 2024).

Far less scrutinized is the alleged underperformance of the police in serving minority civilians when they are *victims* of a crime (Bachman, 1994; Natapoff, 2006; Leovy, 2015). Minority residents of large US cities in fact report being discriminated against when victimized, arguing that police investigations are less thorough when the victim is not White. This is claimed to result in disproportionately many unsolved (i.e., not-*cleared*) crimes and lower levels of accountability for violence committed against these victims, with homicides cited as a leading example (Mueller and Baker, 2016; Tuerkheimer, 2017).[2] This differential treatment can have significant consequences. It can decrease the confidence in the police among minority residents and reduce their future willingness to cooperate, in addition to hindering crime deterrence in minority neighborhoods (Lowery et al., 2018a; Lowery et al., 2018b). Despite the importance of the issue, a rigorous examination of racial discrimination against crime victims is lacking.

In this paper, I fill this gap and test for racial discrimination against crime victims in the context of police investigations of violent crime. I develop a marginal outcome test for racial bias – as opposed to accurate statistical discrimination (Hull, 2021) – against crime victims, derived from a simple model of police investigations. Following Becker (1957), I leverage a post-treatment outcome (i.e., realized upon reaching the conclusion of what the police consider a successful investigation of the crime) to test for biased preferences among investigators. I apply this test to novel data

[1] These disparities extend beyond interaction with the police, as suspected criminals are less likely to be released on bail (Arnold et al., 2018; Arnold et al., 2022) and more likely to be sentenced to prison and for longer (Abrams et al., 2012; Rehavi and Starr, 2014) if they are not White.

[2] Investigation of violent crime is not the only setting where racial discrimination against crime victims might appear. For example, non-White residents of large US cities have reported experiencing slow response times and a lack of attention and care from responding police officers after calling 911 (Sabino, 2024; Mahr and Annie, 2024).

from Chicago, which allows me to examine the behavior of an understudied decision-maker: the police detective. In this setting, the detective has a clear objective: to gather sufficient evidence to ensure a successful prosecution of the perpetrator(s), thereby delivering justice to the victims and their families. Intuitively, the detective will wait to submit a case for court proceedings until the expected success of her investigation has reached an acceptable level, given the evidence collected. Hence, all cases are submitted for prosecution exactly when this desired threshold of evidence, and consequently of success, is achieved.[3]

Empirically, I consider an investigation to be successful if the Cook County State Attorney's Office (CCSAO) – which is the prosecutor of reference for the Chicago police – accepts the case for prosecution. Since this decision is based on the quantity and quality of the evidence presented by the police, it is informative about the detective work (Smith, 2019; Charles, 2022). Importantly, even when the prosecution rejects the case and no one is brought to court, the police can still claim to have solved the crime, ultimately stopping the investigation and classifying the case as an *exceptional clearance*, in contrast to the more standard *clearance by arrest and prosecution*.[4]

I then introduce the first testable definition of racial bias against crime victims in the context of police investigations handled by detectives, which differ in spirit from the one-time interactions typical of patrol officers. Specifically, bias originates from a lower acceptable threshold of expected success upon submission if the victim is not White. Simply put, cases are worked more poorly and are therefore less likely to enter the court process for the group subject to discrimination. These race-dependent thresholds can be micro-founded from a model of police investigation where the detective minimizes type 1 errors (i.e., submitting a case that fails) and type 2 errors (i.e., not submitting – yet – a potentially successful case). It can be shown that biased detectives hold a lower relative cost of type 1 error when the victim is not White.

I perform this marginal outcome test starting from novel data on 820,537 victims of violent crime and the related investigations handled by the CPD detectives from 2001 to 2023. Some striking descriptive facts emerge. Violent crimes are significantly less likely to be solved when committed against Black or Hispanic victims, and even less so when looking more narrowly at the probability of a crime being solved via arrest and prosecution. These gaps persist even after

[3]The model also allows for these thresholds to never be reached, which means that the case is never solved. This occurs if a case is suspended, formally or informally, and this decision can also be affected by the race of the victim.

[4]Another frequent reason for exceptional clearance is when the victim is not willing to cooperate and press charges. I discuss this instance more in detail in Section 2.

controlling for several features of the incident capturing the difficulty of the investigation, restricting to within-neighborhood comparisons, and accounting for the identity of the detective assigned to the case. These disparities, however, need not reflect racial bias: the detectives might be ensuring that all cases have the same likelihood of being accepted for prosecution and enter the court process, therefore accurately predicting the outcome of the next stage in the justice system. Any disparity in acceptance rates would instead point to racial bias. By restricting the focus to cases that *are* reviewed by the prosecution office, we can assess this through the lens of an outcome test.[5]

Violent crimes investigated by CPD detectives are 6.6 pp less likely (-7.25%) to be accepted for prosecution upon submission when the victim is Black, consistent with racial bias defined as a higher tolerance for the failure to deliver justice to Black victims. I find modest evidence of bias against Hispanic victims as their cases are 2.2 pp less likely to be accepted for prosecution than cases of White victims. The main estimates are relatively unaffected by the inclusion of several characteristics of the originating incident and other demographics of the victim (-5.8 pp for Black victims and -1.6 pp for Hispanic victims). This suggests limited scope for bias originating from either other relevant attributes of the victim or the difficulty of the investigation. This also limits concerns about the logical validity of the marginal outcome test (Canay et al., 2023).[6] Cases involving Black victims are less likely to be accepted upon submission across all types of violent crimes in the sample: homicides, robberies, sexual assaults, aggravated batteries and aggravated assaults. Noticeably, racial bias does not vary significantly with the racial makeup of the neighborhood where the crime occured.

Another relevant feature of the crime that should be considered, especially among cases that reach the submission stage, is the identity of the alleged criminal. It is possible that the desired thresholds are a function of the suspect's race as well, and given that crime tends to occur within-race, the detectives' behavior might be driven by the suspect's characteristics, rather than the victim's. The results, however, hold when controlling for the race – and sex and age – of the suspect: cases involving Black victims are less likely to be accepted regardless of the identity of the suspect. If anything, crimes allegedly committed by Black arrestees are 1.9 pp more likely to

[5]As described in the body of the paper, only data from 2006 onward is available for the outcome test.

[6]Canay et al. (2023) shows that models of decision making underpinning outcome tests can be recast as Roy models. However, if the decision-maker follows a generalized Roy model, where unobserved characteristics enter both sides of the decision rule, we might conclude bias even if the decision-maker is unbiased or biased against the opposite group.

be prosecuted than those committed by White suspects.[7] I also document significant differences between dyads of victim-arrestee race: Black on White crimes and White on Black crimes are the most and least likely dyads to be accepted, respectively.

I then tackle potential concerns about infra-marginality of the cases submitted by CPD (Ayres, 2002). If the probability of acceptance of the case does not evolve continuously over time, some investigations might be considered concluded when their expected success has passed the desired threshold, which is particularly likely among very easy cases. If this is more likely to occur among cases of White victims, we could be incorrectly concluding bias among detectives. I address this by leveraging the time elapsed from when the crime was reported to when the case is reviewed by the prosecution. I specifically control for the duration of the investigation, under the assumption that the faster the case has been solved, the more likely the evidence is to lie above the threshold, and the controlling for this quantity would partial out any over-shooting of the thresholds. Nevertheless, I find that *even for cases submitted in the same amount of time*, a rejection is more likely if the victim is Black. Additionally, I drop cases solved very early, which are more likely to be infra-marginal, and the results are qualitatively unaffected.

The outcome of interest for the detective might also not be the *first* stage of the court process, i.e. the acceptance of the case, but the *last* one, i.e. a potential conviction. Using information from CCSAO, I document that cases involving Black victims are 7.4 pp less likely to reach arraignment and 6.7 pp less likely to result in a guilt ruling, even after accounting for the aforementioned controls. Detectives are also more tolerant of a higher risk of violent recidivism for crimes committed against Black victims, and to a minor extent for Hispanic victims. Moreover, consistent with the higher rates of rejections, crime involving Black victims are more likely to be classified as exceptional clearances due to prosecution denial rather than clearances achieved via arrest and prosecution. Regardless of the precise outcome guiding their behavior, detectives close relatively lower quality investigations for cases of Black victims, which is consistent with racial bias against them.

I next explore the drivers behind these results. First, it is possible that the detectives hand the same evidence for all victims to the prosecution, but CCSAO is more demanding for cases involving Black victims. This would mean that the detectives still tolerate a higher likelihood of rejection

[7] The coexistence of these two forces, under-provision of justice services to Black victims and over-punishment of Black suspects has been sometimes referred to as the *paradox of under-protection and over-policing* (Goff, 2021, Boehme et al., 2022, Lewis and Usmani, 2022).

for cases of Black victims, and therefore still display bias, but they do not *directly* set differential standards. Effective policy-making requires knowing which agent it can act upon. Second, racial bias can originate from two sources: biased *preferences* against minority victims (racial animus) (Becker, 1957) or biased *beliefs* held by the detectives about what makes an investigation successful upon completion (inaccurate statistical discrimination) (Bohren et al., 2023). Distinguishing between the two is important in devising the appropriate policy intervention aimed at correcting the detectives' practices.

I show that detectives play an active role in setting race-dependent thresholds by exploiting relevant heterogeneity in the estimates. First, I find bias against Black victims throughout the entire time window, during which three different State Attorneys (SAs) have led the office, ranging from a police-friendly SA with a more orthodox approach to a more recent progressive-minded SA that often clashed with the police department (Conroy, 2001, Spielman, 2015, Grimm, 2022). The consistency of the estimates despite the radical change in prosecutorial approaches over the years suggests that the police preferences, at least to some extent, actively drive investigative behavior and, consequently, the differential rejection rates. Second, estimates of racial bias differ widely across detective areas where crimes are investigated. Since all areas interact with the same State Attorney's office, and cases are randomly assigned to individual ASAs for review, this is evidence that the police's biased preferences matter. Third, and in the same spirit, detective-level estimates of bias, obtained through a value-added approach, reveal substantial heterogeneity that cannot be explained by the tendencies of the ASAs who conduct the review.

I then try to disentangle between biased beliefs and biased preferences by leveraging detectives' characteristics, similarly to Arnold et al. (2018). I find small and statistically insignificant interactions between the experience of the detective who clears the case and the race of the victim, across several definitions of experience. This suggests that as detectives gain experience and, presumably, learn on the job, adjustments in beliefs do not change their differential submission standards.[8] Ultimately, I can rule out that the main results are driven by inaccurate beliefs, and detectives indeed display racial animus against Black, and to a lesser extent Hispanic, victims. I then test whether biased preferences are a function of the detective's race. A common assumption in the discrimination literature is that, if biased preferences are indeed behind the observed disparities, the rankings of the outcome by race of the victim, or at least the size of the gaps, would depend

[8] Nonetheless, additional experience increases the baseline probability of acceptance across all races.

on the race of the decision-maker (Anwar and Fang, 2006; Antonovics and Knight, 2009). In my setting, I do not find any evidence of race concordance between victims and detectives. While biased preferences remain a more credible explanation than biased beliefs, they do not depend on the detective's race. On the other hand, I find suggestive evidence that biased preferences among detectives are consistent with the views of the public and/or scrutiny from the media: Black (homicide) victims, and to a minor extent Hispanic (homicide) victims, are less likely to receive media attention from the main local newspapers in Chicago, even after controlling for other features that determine the newsworthiness of the incident. Moreover, heterogeneity in media coverage by victim profile reproduces the same patterns of racial bias I document using the outcome test. Although descriptive in nature, these disparities in media coverage are consistent with reduced pressure on the detectives to achieve successful case outcomes for Black victims, translating into a lower relative cost of type I errors and weaker submissions.

My findings suggest that transparency initiatives in policing, which focus on areas such as traffic stops and use of force, should also address police investigations and disparities in the treatment of crime victims more broadly. Additionally, using clearance rates as a metric for police performance is likely inadequate to evaluate the deterrent power of policing. Many cases considered solved in fact involve arrests that never reach court proceedings, and they do so differentially by victim race.

Related Literature: I contribute to the literature on discrimination against minority individuals in the US justice system. A rich body of work in economics has studied the overly punitive tendency of the US justice system, and the police in particular, towards minority individuals. Most such work has focused on racial discrimination against individuals who are *suspected criminals*. Few papers have instead considered the treatment of crime *victims*. An exception is Alesina and La Ferrara (2014), who tests whether Black defendants are more likely to be sentenced to death than White defendants by studying whether the ranking of error rates by defendant race changes with the race of the victim. Their main question, however, remains whether Black defendants are over-sanctioned. Moreover, they cannot identify marginal individuals, requiring strong assumptions on the distribution of the unobservables.[9] Criminologists have devoted more attention to the treatment of victims, finding mixed results (Taylor et al., 2009; Petersen, 2017; Fallik, 2018). Most

[9]Another exception in economics is Bjerk (2022), which descriptively discusses differential patterns of clearance by victim and neighborhood, albeit in the context of a different research question.

of these studies leverage data voluntarily reported by police agencies to the FBI (NIBRS) and compare the probability of clearance across victim races while controlling for factors that are thought to correlate with the difficulty of the case. Using these data and adopting a correlational approach raises identification concerns.[10] Moreover, this literature has overlooked the identity of individual detectives and the link between investigations and subsequent stages of the justice system. I attempt to address these limitations. First, I define discrimination within a model of investigative behavior to guide identification. Second, I leverage data on individual detectives. Third, I use a court-related outcome realized upon crime solution to elicit decision-makers' preferences.

I also contribute to the literature studying police behavior broadly defined. Most papers examining individual decisions made by the police have focused on the behavior of patrol officers. However, little is known about another crucial figure in law enforcement: the police detective. Detectives represent a relevant fraction of police employees and, importantly, significantly affect the probability of apprehension of criminals, determining the deterrent power of policing.[11] Economists often tangentially analyze the work of detectives when treating clearance rates as proxies for a police department's productivity, almost always as aggregates.[12] To the best of my knowledge, this is the first paper that focuses on the individual decisions made by these agents and how they vary by the race of the victim involved in the assigned case.

Methodologically, I contribute to a large literature that develops methods to measure discrimination. A long tradition in economics has used outcome tests, which are based on differences in post-decision outcomes across groups, to do so (Becker, 1957; Becker, 1993). A critical issue in this context is to identify individuals on the margin of being treated, as opposed to using group averages, which pool infra-marginal observations as well.[13]. A common solution is to impose strong distributional assumptions on the unobservables of the individuals involved (Knowles et al., 2001; Anwar and Fang, 2006, Alesina and La Ferrara, 2014). Alternatively, recent work has instead leveraged quasi-random assignment to decision-makers to identify marginal individuals (Arnold et al.,

[10] First, reporting to NIBRS is voluntary, poor and skewed towards smaller jurisdictions. This makes the results from these studies representative of medium- or small-sized cities, where policing and crime are radically different. Additionally, any null effect might be biased due to selection on consistently reporting to NIBRS. Moreover, these studies control for features that are either revealed only upon clearance or that directly affect investigative intensity, as noted by Cook and Mancik (2024).

[11] Of sworn officers employed by agencies with at least 100 officers, 27% reported working as investigators (Prince et al., 2021).

[12] For examples, see McCrary (2007), Mastrobuoni (2020), and Mastrorocco and Ornaghi (2020).

[13] This is the so-called *infra-marginality* problem Ayres (2002).

2018; Dobbie et al., 2021). In the absence of quasi-random assignment, I circumvent the issue and identify racial bias by exploiting the dynamic nature of police investigations. Detective operations are concluded precisely when a desired threshold of evidence is reached, making all the observed solved cases on the margin of being closed.[14]

Outline: The remainder of the paper proceeds as follows. Section 2 provides relevant background on how police clear a crime and the diverging paths a cleared case can take, especially in Chicago. Section 3 describes the main data used in the paper. Section 4 sketches a model of investigations and derives the marginal outcome test for racial bias in investigations. Section 5 presents the main results from the marginal outcomes test. Section 6 analyzes the sources of the observed bias. Section 7 discusses the policy implications and concludes.

2 Background

2.1 Crime Investigations in Chicago

Once a crime is reported to CPD, patrol officers are dispatched for the initial response. They conduct a preliminary investigation of the scene, provide assistance to the victim, interview witnesses, and apprehend any suspects who are still present (Chicago Police Department General Order G04-01). If the type of incident requires immediate notification per CPD Directives, the officers contact the respective Detective Area (or Police Area), which then dispatches detectives from the appropriate section (Chicago Police Department Special Order S04-21-01).[15] When on the scene, detectives from that detective area take over the investigation, processing physical evidence and interviewing persons having knowledge of the crime. If required, they are assisted by forensic technicians as well (Chicago Police Department General Order G04-02). Detective Areas are geographic partitions of the city where detectives are stationed. They encompass several districts and represent the level at which detectives are assigned to cases. For budgetary reasons, CPD reduced the number of these

[14] Other work in the realm of crime and justice cleverly exploits bunching points in the relevant decisions to identify discrimination (Tuttle, 2019, Goncalves and Mello, 2021). Another common solution is to condition on rich observables in an attempt to compare individuals of different races who are otherwise as similar as possible (Fryer Jr, 2019, Lieberman, 2024). I adopt this latter approach in Section 3 when studying the extensive margin of discrimination but move to a model-based identification from Section 4 onward.

[15] Each area had a Violent Crime Section and a Property Crime Section until 2020. In early 2020, CPD introduced a citywide Homicide Section and added homicide sections in each area.

areas from five to three in March 2012, then restored them to five in April 2020 (Byrne, 2020). See Figure A.2 for the corresponding maps.

Detectives, like patrol officers, are assigned to one of three shifts. Case assignments tend to follow a rotation system, but only to some extent. In practice, assignments often deviate from quasi-random allocation within shifts, as detectives with particular knowledge of certain neighborhoods or types of incidents are more likely to be assigned those cases.[16] The detectives dispatched to the scene are not necessarily those who conduct follow-up investigation as primary on the case.[17] Table A.1 confirms this by showing that even conditional on area-by-time-by-crime type fixed effects, victims observe their cases being assigned to detectives belonging to the same race or ethnicity. This correlation persists even when controlling for other characteristics of the crime. Cases involving Black victims are, if anything, assigned to detectives with lower clearance rates than those assigned to cases of White victims.

2.2 How to Solve a Crime

According to FBI terminology, a solved crime corresponds to a "clearance", and 2 main types of clearance are typically considered (FBI, 2010).[18] First, a law enforcement agency can clear a crime by successfully arresting a suspect, which represents the vast majority of clearances. This occurs when the police have identified, arrested and turned over for prosecution at least one suspect. An alternative way of clearing a crime is by exceptional means. This occurs when the police have identified a suspect, gathered what they deem is sufficient evidence to support arrest and prosecution of the suspect, but encountered a circumstance beyond their control that prohibits them from doing so. Examples of these exceptional circumstances are victims' refusal to cooperate,

[16]The lack of quasi-random assignment prevents me from using, among others, IV methods such as those in Arnold et al. (2018) or an "identification at infinity" approach as in Arnold et al. (2022)). Even if cases were quasi-randomly assigned to detectives, the exclusion restriction is unlikely to hold. Unlike bail judges, detectives handle several decisions and therefore not only determine *which* case is solved, but also *how* they are solved - and viceversa. IV methods would additionally require a first-stage monotonicity assumption, which has come under greater scrutiny recently (Mogstad et al., 2021, Arnold et al., 2022).

[17]A review written in 2019 by the Police Executive Research Forum (PERF) on CPD's investigative practices, per request of CPD and funded by DOJ, offers more details on homicide and shootings assignments. An instance of re-assignment is the "longstanding practice of midnight (or 1st watch) detectives transferring their cases to dayshift detectives. For homicide cases, these detectives typically respond to the scene, but then hand off the case to a dayshift detective for follow-up investigation" (PERF, 2019).

[18]The FBI defines these two types of clearance, and the circumstances leading up to them, mostly from a data collection perspective. In reality, law enforcement agencies sometimes deviate from these guidelines.

the prosecution's refusal to press charges or the death of the identified offender.[19] In essence, for the police to clear a crime, it often suffices to identify a credible suspect, regardless of whether they can apprehend and/or prosecute the individual.

Whether a crime is solved depends both on police behavior and incident-specific external factors, which determine the intrinsic difficulty of the investigation. For example, firearms leave less evidence than other weapons involving interpersonal contact, such as knives (Asher, 2021). Cooperation from witnesses (and/or victims if not a homicide), or the lack of thereof, is another influential factor that police report to strongly affect the likelihood of solving a crime (Asher, 2021). The actions police can take during a criminal investigation are numerous and multifaceted, ranging from the number and manner of interviews to the thoroughness of evidence collection and lab submissions, as well as persistence in follow-ups with witnesses, families, and victims.

2.3 At the Conclusion of the Investigation

While practices might vary by jurisdiction, whenever a suspect is found and is detained and the victim is willing to cooperate, the police typically decide whether to release her without charges or request that the prosecution press charges against her. In Cook County, the relevant jurisdiction for CPD, if the police opt for the latter, they contact the Felony Review Unit (FRU) at the CCSAO. This is typically a rotating group of Assistant State Attorneys (ASAs) who review the police's evidence and decide whether to approve or reject the charges against the suspect in custody (Cook County State's Attorney, 2018, Cook County State's Attorney's Office, 2024). The police contact the FRU for felonies, while they can bypass the FRU for misdemeanors and drug charges (Chicago Police Department Special Order S04-21-01). The contacted ASA can decide whether to accept the charges based on the evidence presented, reject the charges or suggest that the investigation continue.[20] Whenever charges are not accepted, the police release the suspect but can resubmit for charges at a later time. This is an instance when the police can classify the case as an exceptional clearance, ultimately stopping the investigation without any suspect going to court. The precise clearance classification is up to the police and does not necessarily perfectly match the result of the

[19] According to calculations made by Kaplan (2021) using 2019 NIBRS numbers from all agencies that reported data, approximately 50% of all exceptional clearances are due to a lack of cooperation from the victim, and 50% are due to the prosecution denial. These numbers mostly reflect trends for non-deadly crime, as homicides represent a very small share of total crime.

[20] From discussions with the CCSAO, continued investigation is suggested when there is a clear piece of evidence that is missing and is considered obtainable.

felony review, especially when multiple suspects are involved in the review.[21] Whenever charges are accepted, the suspect(s) will be scheduled for a preliminary hearing before potentially appearing at arraignment, where they are formally charged, and going to trial and sentencing. If the case is accepted, it is typically classified as a clearance via arrest and prosecution, regardless of the outcome of the court process. Figure A.1 shows the arrest-to-court pipeline in Cook County, IL. The outcome test developed in this paper focuses for the most part on the first step, when after an arrest the police go to the CCSAO's FRU.

From a legal standpoint, a discrepancy between police work and the prosecution's opinion can arise because the standards for arrest and prosecution need not coincide. The law in fact requires that the totality of the evidence amounts to the level of *probable cause* for the police to be able to arrest and request charges on a suspect. However, for the prosecution to be able to convict the suspect(s), the threshold increases to *proof beyond a reasonable doubt*.[22] It is therefore at the discretion of the police whether to gather evidence exceeding probable cause before submitting a case for prosecution or they deem probable cause to be sufficient. Analogously, the prosecution can accept cases that are not (yet) backed by evidence amounting to proof beyond reasonable doubt. However, given the incentive on their part to maximize conviction rates, it is in their best interest to reject a case whenever they deem the evidence insufficient. The FRU's decision is therefore informative of the quality of the police work: acceptance of charges means that the CCSAO deemed the evidence to be good enough to seek a conviction, i.e., at the level of proof beyond reasonable doubt. Rejection can instead be interpreted as the CCSAO considering the evidence only at the level of probable cause. Note that even if charges are rejected, the police can consider the crime to be solved. In this case, the crime is classified as cleared by exceptional means. Upon rejection, the police decide whether to continue investigating and again seek charges or stop completely. The non-negligible rate of rejections by the CCSAO has been repeatedly highlighted

[21] Police agencies in the US handle exceptional clearances in different ways. Spohn and Tellis (2010) point out that a through interpretation of the FBI guidelines would entail cases being classified as exceptional clearances only when arrests are physically impossible. Many agencies, CPD included, adopt a laxer interpretation. Additionally, CPD and CCSAO seem to disagree on who governs the decision to exceptionally clear a crime. Police claim that it is the prosecutorial rejection that determines it, while CCSAO believes that the clearance classification is purely up to the police and ASAs solely opine on the quantity and quality of the evidence Riedel and Boulahanis, 2007.

[22] Probable cause means that, based on the existing evidence, a reasonable person would believe that an offense occurred and that it is probable that this suspect committed that offense. On the other hand, proof beyond reasonable doubt also requires that there is no reasonable doubt that the defendant committed the crime. For reference, see the US Supreme Court rulings in Brinegar v. United States, 338 U.S. 160 (1949) and Maryland v. Pringle, 540 U.S. 366 (2003). Probable cause and proof beyond reasonable doubt are not to be confused with *reasonable suspicion*, which is an even lower standard that is grounds for an investigative stop by law enforcement.

by the media as a problematic issue that can fuel a cycle of violence, especially in the context of murder investigations (Smith, 2019; Charles, 2022; Grimm, 2022).

3 Data and Descriptive Facts

3.1 Data

Through a series of FOIA requests, I obtained data on all victims of violent crime reported to the CPD from 2001 to mid-2024 and the related investigation. The data include information on both fatal violent rime (first-degree murder) and non-fatal violent crime (aggravated assault, aggravated battery, criminal sexual assault, robbery). I have information regarding the victim of the incident (race, ethnicity, gender, age), the incident itself (type of crime, weapon used, premises of incident, address, date and time, circumstances of the incident), all the detectives assigned as primary to the case, including their demographic characteristics (race, gender, date of appointment as police officer) and the date of assignment to each case, and the status of the investigation (whether the crime was cleared and the type of clearance, if cleared). The status allows me to distinguish between 3 major categories of clearances: clearances achieved via arrest and prosecution, exceptional clearances due to the victim's refusal to cooperate and exceptional clearances for other reasons. The latter pools exceptional clearances due to prosecution denials and due to death of the offender. Since only a small of exceptional clearances occur due to the death of the offender, this third category mostly represents cases that are not prosecuted, as confirmed by communications with CPD (Kaplan, 2021).[23] The status of the investigation does not always match the result of the felony review, among those cases that are reviewed. For this reason, the outcome test at the core of the paper will be based on the more precisely recorded result of felony review rather than the overall status of the investigation.

I drop all offenses that occurred in 2024 to allow some time for investigation. Unless specified, I keep only victims that are either White non-Hispanic (henceforth, White), Black or White Hispanic (henceforth, Hispanic), and I restrict the sample to cases that have ever been assigned to a detective. [24] These two restrictions leave me with 751,999 victims. The dataset is kept at the victim-level

[23]For homicides, on the other hand, I can isolate cases cleared due to the death of the offender. This is because homicide-related information is recorded by CPD in a separate, more detailed database. Cases that are explicitly rejected for prosecution are instead labeled as "Bar to prosecute".

[24]Not all violent crimes have a primary detective assigned. These incidents are typically resolved on the scene by

since there is variation in demographics, mostly in age and gender, and features of the crime even within incidents.[25] Although there can only be one primary detective at a time assigned to a case, they are sometimes replaced during the course of the investigation due to a variety of reasons (arbitrary decision of the supervisor, retirement, vacation, etc.). When using the full sample, I keep the first detective ever assigned to maintain consistency between solved and unsolved cases. When conducting the outcome test, which restricts to cases that are solved, I keep the detective that was on the case at the time of review of the case.

I also obtained data on all adults arrested by CPD employees from 2001 to 2024, which can be matched to the incident using a common identifier.[26] This data provides information on the race, gender, age at time of the incident of the arrestee, time and location where the arrest took place, a list of all charges brought against them, with related type and class. I keep only arrestees who are either White, Hispanic or Black to match the filter imposed on crime victims. Importantly, the data contains information on felony review for each arrest that requires so. Namely, I observe the decision taken by the FRU and date and time of the review. This allows me to restrict the sample to those crimes that are linked to arrests that resulted in felony review. Unfortunately, information on felony review started to be systematically collected from 2006 onward. Therefore, the sample used for the outcome test is restricted to all adult arrests, involving a felony, made from 2006 to 2023 in connection to a violent crime committed anytime between 2001 and 2023. When using arrestee information, I expand the dataset to the victim-arrestee level and weight each observation by the product of total victims and total arrestees linked to the given incident.

Finally, I add court information using data obtained by the Chicago Data Collaborative from the Circuit Court of Cook County, which records information from arraignment to sentencing for each arrestee who reaches such stages. This data covers all court proceedings from 1984 to mid-2019, I further restrict the sample to felony arrests reviewed from 2006 to 2016 enough time for court proceedings to ensue.

the responding officers and do not require any follow up, or are more minor crimes that do not warrant a follow-up investigation (Special Order S04-21-01). Roughly 70% of the incidents without detectives assigned are cleared, and mostly via arrest and prosecution. The remaining 30% are for the most part assaults without injuries involved.

[25]More precisely, the dataset is at the victim-incident level. I cannot follow victims across incidents.

[26]Data on juvenile arrests is not shared by CPD for privacy reasons.

3.2 Descriptive Statistics

Table 1 reports summary statistics for the full sample, separately for White, Black and Hispanic victims. Since homicides are vastly outnumbered by other types of violent crimes, but are also highly salient in the public minds, I keep them separate from non-fatal violent crimes. There is a large gap in the raw probability of clearance between White and minority victims of homicides. Cases of minority victims are also less likely to be solved via arrest and prosecution, even conditional on clearance. For non-fatal violent crime, in contrast, we observe a relatively small overall clearance gap which now seems to favor Black victims; however, this gap becomes more substantial and reflects the ranking observed for homicides if we consider clearances that occurred via the successful arrest and prosecution of a suspect. Consistently, refusal to cooperate is also most common among Black victims. This shows how the overall clearance rate can be a misleading measure of how often crimes get solved: only between slightly more than half of the victims whose cases are labeled as solved observe their offender going to court.

White victims of homicides are more likely to be female and older, and they represent a small share of total victims of homicide. Robberies are by far the most common of the non-fatal violent crimes, especially for White and Hispanic victims. Domestic violence is more prevalent among White victims if the incident ends up in a homicide, while the opposite is true for non-fatal violence. Firearm violence is more common among minority victims, especially for deadly acts of violence, where the share of firearm-related homicides exceeds 80%.

Table 2 shows the characteristics of the detective in my sample. Roughly two-thirds of the detectives working for CPD between 2001 and 2023 are White, Black and Hispanic detectives represent approximately 16% of the total each, and most of them are men. The average detectives has 84% of her cases belonging to the same type of crime, but not even half of the total are hyper-specialized, which is consistent with anecdotal evidence (PERF, 2019).

Figure 1 shows the demographic distribution of neighborhoods, defined as Census tracts, in Chicago with the clearance rate. In only a very small share of Census tracts more than half violent crimes are solved and in the vast majority of neighborhoods less than one-third of these crimes are solved by arrest and prosecution. Figure 2 provides a more accurate picture of the relationship between race and the outcome of the investigations, by showing how clearance rates change as the share of minority residents in the Census tract increases. There is an overall negative relationship when considering the overall clearance rate (Panel a), which however hides more nuanced trends. On one hand, if a crime is solved, the likelihood of this occurring via arrest and prosecution also decreases as the share of White residents decreases (Panel b). On the other hand, I document the opposite relationship for crimes solved due to the victim's refusal to cooperate, conditional on the crime being solved (Panel c). All of these patterns become particularly stark in the top decile of the share of minority residents. Figure 2 illustrates further how the overall clearance rate is an imperfect measure of crime solution and accountability, especially when examining racial patterns.

3.3 Observational Comparisons

A natural question is whether differences in the probability of clearance persist once we compare very similar crimes committed against White and minority victims. This is an approach often adopted in the discrimination literature when investigating the disproportionate selection into an adverse treatment by a potentially biased decision-maker (Rehavi and Starr, 2014, Fryer Jr, 2019, Lieberman, 2024). The analysis in this setting, however, is complicated by the fact that we can identify four, rather two, major outcomes of an police investigation: 1) unsolved; 2) cleared via arrest and prosecution; 3) cleared by exceptional means due the victim's refusal to cooperate; 4) other exceptional clearances, with the vast majority represented by cases when the prosecution refuses to press charges. Exceptional clearances are especially difficult to classify, as they are not obviously successful or failed investigations.

I still test for residual differences in the probability of clearance by race of the victim after controlling for several other features of the crime. I use 3 different definitions of clearance: *i)* overall clearance (cases belonging to categories 2), 3) and 4); *ii* clearances *not* achieved due to the victim's refusal too cooperate cases belonging to categories 2) and 4); *iii)* clearances achieved explicitly via arrest and prosecution (cases belonging to category 2)).

I estimate the following model:

$$D_i = \alpha + \beta^B Black_i + \beta^H Hispanic_i + \gamma' X_i + \delta_j + \delta_{ayc} + \varepsilon_i \tag{1}$$

where D_i is the observed clearance status of the case for victim i, which takes value 1 if the case has been solved. $Black_i$ and $Hispanic_i$ take value 1 if the victim is Black or Hispanic, respectively. The vector X_i includes a set of characteristics of the victim and the incident other than the race of the victim that might correlate with it. These include: sex and age (binned in 6 categories) of the victim, whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. I include fixed effects for the first detective assigned to the case (δ_j), and detective area-time-crime type fixed effects (δ_{ayc}). Time is defined as the specific month-year combination when the incident occurred, and there are five major types of crime (homicides, aggravated assaults, aggravated batteries, robberies, sexual assaults). I also progressively add controls for the neighborhood where the crime occurred. First, I include fixed effects for the quartiles of share of Black and Hispanic residents, separately, and the median income in the tract, logged to capture non-linearities.[27] Second, I replace these controls with tract fixed effects, which hold any time-invariant characteristics of the neighborhood fixed. I weight each observation by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level.

As shown in Table 3, Black and Hispanic victims experience significantly lower clearance probabilities, regardless of the precise definition of clearance. Violent crimes involving Black (Hispanic) victims are roughly -3 pp (-2.4 pp) less likely to be solved when adopting a broad definition of clearance. For Black victims, these gaps increase in magnitude, to -4.2 pp, when not considering exceptional clearances due to the victim's refusal to cooperate as solved. The difference in clearance probability slightly shrinks to -3.8 pp when narrowing the definition of clearance to cases solved via arrest and prosecution. For Hispanic victims, the estimates progressively decrease to -1.9 pp and then -1.3 pp. Disparities by characteristics of the neighborhood emerge as well, independently of the victim's race. Clearance rates decrease once the share of Black residents exceeds 50%, while the gradient of the share of Hispanic residents is more monotonic. There is also evidence that median wealth in the Census tract positively affects the probability of clearance.

For these estimates to reflect discrimination, the included controls would have to perfectly capture the difficulty of the investigation, which can be considered in this setting the appropriate

[27] I use the 5-years estimates from the 2010 American Community Survey. While the sample starts from 2001, estimates from the 2000 Census would be less representative of the racial makeup and income composition of the Chicago neighborhoods at the end of the 2010s and beginning of the 2020s.

measure of *qualification* to the treatment, with the treatment being clearance (Bohren et al., 2022). Even if that were the case, it is not straightforward to classify the four potential outcomes of an investigation as desirable or undesirable.[28] In the sample used for the outcome test developed in the next section, each case has only two possible outcomes - acceptance or rejection, which map more directly to successful and unsuccessful. In addition, difficulty is an issue for identification in the outcome test only insofar as it affects the thresholds of evidence that detectives seek before concluding the investigation. Although difficulty almost certainly affects *whether* a case is solved, it is less obvious that it influences *how well* the case is closed, *conditional on the case being solved*, also given the legal standards provided by the law for this margin. The outcome test developed next rests on the baseline assumption - still relaxed later on in the paper, that difficulty determines how fast the detectives reach their ideal stopping point, but not the stopping point itself.[29]

4 Model and Marginal Outcome Test

I next sketch a model of police investigations that allows me to develop a marginal outcome test for racial bias. Following Becker (1993), I leverage a post-decision (i.e., post-conclusion of the investigation), downstream outcome to elicit the agents' preferences.[30] The model explicitly considers two key decision variables that can lead to a clearance. These are *i)* the time when the detective desires to clear and close the case; *ii)* the time when she would instead suspend altogether the investigation of a case that would therefore never be solved.[31]

[28] This is especially true for exceptional clearances due to a victim's refusal to cooperate: some instances may reflect the victim's preferred resolution, while others may result from inadequate communication, coordination, or effort by the police (Klibanoff and Ryan, 2018, Paul, 2016). The marginal outcome test developed in Section 4 addresses these challenges by restricting attention to cases that reach the prosecution's office, whose victims are willing to cooperate. Less than 2% of these are classified as "solved due to the victim refusing to prosecute", and even among these cases approximately 75% are accepted.

[29] As argued in Section 2, detectives need at least probable cause to make an arrest, but they are aware that the prosecution requires proof beyond reasonable doubt to win a case in court.

[30] Post-treatment is slightly inaccurate for this setting since the outcome is realized at the time of clearance: once the case is ready to be closed, the outcome is also realized.

[31] The test derived in this paper is similar to those in Anwar and Fang (2015) and Lodermeier (2023). They leverage the evolution of a relevant variable over time and use the (first) period when a decision is made by the relevant agent to identify marginal cases in the parole and eviction settings, respectively. Lodermeier (2023), in particular, does so without imposing strong parametric assumptions and is the closest in spirit to my approach.

4.1 A Model of Police Investigations

Setup: The police receive case i, which is then assigned to an investigator. The investigator observes the race $RV_i \in \{w, m\}$ of the victim, as well as other characteristics U_i that are in part unobserved to the econometrician. The investigator chooses how many days T_i to keep the case open and work on it before submitting it for clearance and, ultimately, for prosecution of the arrestee(s). She has the option of submitting the case on any day t following a decision rule that tells her whether the case is ready for submission. Let $D_i(t) = 1$ denote the investigation being concluded and the case being submitted for prosecution – at time t.

Upon submission, the case can be successful or not, and this partly depends on how much time was spent on the case. As a measure of success, I use the DA's decision to accept the case and proceed to prosecution. Let $Y_i^*(t) \in \{0, 1\}$ indicate the potential success when days $T_i = t$ are spent on the case before submission, and write the observed outcome as $Y_i = Y_i^*(T_i)$. The true probability of success upon submission of the case at the end of each period t is $p(t, rv, u) = E[Y_i^*(t) \mid RV_i = rv, U_i = u, T_i = t]$. To simplify notation, expected success is denoted as $p(t)$. I assume that $p(t)$ is weakly increasing in t, all else fixed: the more time is spent on the case, the more evidence is collected, and the more successful the case is going to be in expected terms. I further assume that $p(0) = 0$ and that time is continuous.

The investigator also pays a subjective cost $c(RV_i)$ of submitting a case with a given evidence level to court. I allow the cost to depend on race, with $c(w) \neq c(m)$ signifying racial discrimination. The lower $c(RV_i)$ is, the more the detective is willing to submit at low values of $p(t)$, i.e., with worse evidence. The submission cost therefore identifies the desired evidentiary standard that has to be met before submission a victim of race R_i. The cost $c(RV_i)$ can be further micro-founded by assuming that the detective minimizes the costs of type I errors (submitting potentially failing cases) and type II errors (not submitting – yet – potentially successful cases). $c(RV_i)$ can be then recast as the relative cost of type I errors, i.e., submitting a potentially failing case, for a victim of race RV_i.[32] Therefore, if she discriminates against minority victims, we expect $c(m) < c(w)$: failures are relatively less costly if the victim is not White.

Clearance: The investigator's desire to clear, submit and close the case can then be represented

[32] See Appendix B.2 for the derivation.

as follows:

$$D_i(t) = \mathbf{1}[p(t) \geq c(RV_i)] \tag{2}$$

There exists a period t_i^* satisfying the indifference condition $p(t_i^*) = c(RV_i)$, at which point the investigator desires to submit and close the case. This is equivalent to the investigator having race-specific thresholds of minimum expected success that she is willing to tolerate before submitting, $\underline{p}(w)$ and $\underline{p}(m)$. If she has a distaste for investigating cases of non-White victims, we expect a lower probability of success for non-White cases filed on the margin than White cases filed on the margin, i.e., $\underline{p}(m) < \underline{p}(w)$.

Note that these thresholds are assumed to depend on the race of the victim, but not on the other characteristics of the case, which jointly determine the difficulty of the investigation. These characteristics in fact enter the left-hand side, but not the right-hand side. This means assuming that they affect the pace at which evidence accrues and how fast the case reaches the desired quality, but not the desired thresholds of expected success themselves. In the empirical analysis, I relax this assumption and test whether any differential standards that I might find vary with observables other than race but potentially correlated with it. This assumption also preserves the logical validity of the marginal outcome test, a point stressed by Canay et al. (2023).

Suspension: Not all cases, however, will reach submission and clearance. The investigator can also suspend the investigation at any point in time even before submission if she considers the investigation not worth being kept open, by either formally or informally abandoning the case. This can be captured by the investigator also paying a subjective cost for actively investigating the case on day t, $l(RV_i, U_i, t)$, which is potentially race-dependent and is strictly increasing over time. If she discriminates against non-White victims and dislikes investigating their cases, we expect $l(m, u, t) > l(w, u, t)$: she finds spending time on cases of non-White victims more costly than investigating cases of White victims. [33] If the investigation is suspended, the case will not be cleared or brought to court. Let $S_i(t) = 1$ denote the investigation being suspended at time t. The investigator will do so if the cost of investigation at time t exceeds the maximum effort that she is

[33] While the suspension decision could be modeled in even greater detail, the decision is kept as simple as possible due to the lack of appropriate outcomes and data for an outcome test. Here, for example, I assume that the detectives know their cost of investigation. However, we can relax this assumption and make $l(RV_i, U_i, t)$ an expected cost: $l(RV_i, U_i, t) = E[l(t)|RV_i, U_i]$.

willing to exert on a case, $\bar{l}$:

$$S_i(t) = \mathbf{1}[l(RV_i, U_i, t) \geq \bar{l}] \tag{3}$$

There is then a period t_i^0 such that the indifference condition along this margin holds, i.e., $l(RV_i, U_i, t_i^0) = \bar{l}$. Equation 3 plays then the role of a (subjective) resource constraint forcing the detective into a corner solution, with the investigation not resulting in a clearance. While perhaps simplistic, Equation 3 is meant to capture the fact that the investigation will be suspended if it is considered unproductive and unlikely to reach the desired threshold of evidence/success for submission.

A case is therefore cleared and submitted if there is both enough evidence to do so, according to the investigator's subjective thresholds of success, as pinned down by $c(\cdot)$, *and* the investigation was never abandoned. The realized time spent on the case is then either the time of suspension or the time of clearance, depending on whether the case was ever suspended, i.e., $T_i = S_i t_i^0 + (1 - S_i) t_i^*$. Clearance status will be equal to 1 only if the case was never suspended: $D_i = (1 - S_i)$. Success will always be equal to 0 if the case is never solved/submitted: $Y_i = D_i Y_i^*(T_i) = (1 - S_i) Y_i^*(T_i)$. Cases that are suspended will not arrive at the DA's office, and we therefore never observe their potential success.[34]

A test for racial bias: In this model, there is room for discrimination along two margins: the investigator could both *i)* assign a lower chance of clearance to cases of Black victims by suspending potentially workable cases – discrimination on the extensive margin – and *ii)* set lower desired success rates even conditional on clearance/submission for cases of minority victims – discrimination on the intensive margin. Note that suspension and clearance here follow two decision processes that evolve in parallel.[35]

While the suspension decision does not involve a downstream potential outcome that is linked to the treatment, the submission decision does. Clearance is in fact determined once the expected downstream outcome Y_i, the successful submission for prosecution, is optimized relative to the costs. Moreover, all cleared cases are closed at the time of indifference, t_i^*. We can therefore perform a marginal outcome test for racial bias on the intensive margin by studying cases that are cleared and their success probabilities once they reach the SA's office. The observed disparity in success rates

[34]Note that the suspension decision will affect which cases are submitted. Case i is not reviewed if $t_i^* > t_i^0$, making Δ^Y representative not of all investigations but only those that are brought to the end.

[35]This setting is close in spirit to use of force by the police: officers both decide whether to use force, potentially in a discriminatory way (Fryer Jr, 2019; Hoekstra and Sloan, 2022), and what *level* of force to use conditional on using force, also in a potentially in a discriminatory way (Lieberman, 2024).

between cases involving non-White and White victims therefore identifies the difference in costs of submission, i.e., desired evidentiary standards, $c(rv) - c(rv')$. Since we defined this difference as racial bias, and cases are closed on the margin, we can state the following:

Proposition 1. If $\Delta_t^Y \equiv E[Y_i \mid RV_i = m, D_i = 1] - E[Y_i \mid RV_i = w, D_i = 1] < 0$, the investigator displays racial bias on the intensive margin against non-White victims.

We would therefore conclude that the police display bias against non-White victims if we estimate $\Delta^Y < 0$. [36] For the proof of Proposition 1, see Appendix B.1.

4.2 Empirical Approach

I next describe how I empirically estimate Δ^Y. I restrict the sample to cases that I know went through FRU and estimate the following baseline model:

$$Y_{ik} = \alpha + \beta^B BlackVictim_i + \beta^H HispanicVictim_i + \delta_j + \delta_{ayc} + \varepsilon_{ik} \tag{4}$$

where Y_{ik} is a dummy that takes value 1 if the case of victim i and arrestee k is accepted by the FRU. δ_j are detective fixed effects, and δ_{ayc} are detective area-month-year-crime type fixed effects. Detective area fixed effects are included because investigative operations (e.g., detective assignment, supervisor review) are conducted at this geographic level, and they can differ widely in their internal functioning and personnel. I interact these fixed effects with month-year fixed effects for when the case was reviewed, as the boundaries of the areas changed substantially over the years. This also accounts for changes in the composition of the pool of ASAs reviewing the cases and the State Attorney herself. The coefficients of interest are β^B and β^H, which estimate total racial bias in closure decisions, against Black and Hispanic victims, respectively. I weight each observation by the inverse of the product of the total number of victims and total number of arrestees involved in the incident.[37] The basic model is then augmented with additional fixed effects and controls to explore how the gap in success rates, if any, changes once we account for

[36] Note that I follow Hull (2021) and allow bias to arise via non-race characteristics.

[37] This makes the target parameter a case-average, down-weighting many-arrestee and many-victim cases symmetrically. Although the outcome is set at the arrestee level, I choose this weighting scheme since when multiple arrests are involved in the same case, they are typically correlated, potentially inflating the role of each case.

other factors that potentially correlate with race. Standard errors are clustered at the detective area-month-year-crime type level.

4.3 Sample for the Outcome Test

It is useful to remind the reader of the sample restrictions imposed for the outcome test. The sample is made of all adult arrests involving a felony that are made by CPD employees between 2006 and 2023 for violent crimes committed between 2001 and 2023. While also crimes that do not involve felonies can be dropped by the prosecution down the line, this can occur also because these are by definition minor crimes. Adopting the sample restrictions described here ensures that we consider only crimes that are reviewed immediately after the conclusion of the investigation and that are dropped due to lack of evidence and not for ethical concerns regarding the gravity of the crime.

As discussed in Section 2, the outcome of the felony review does not always align with the classification of the investigation, which is set at the case level and is common across all potential arrestees. Specifically, cases rejected by the FRU are not always recorded as exceptional clearances, and cases accepted by the FRU are not always recorded as clearances by arrest and prosecution. This mismatch arises for several reasons. First, when multiple arrestees are involved in a review and receive different decisions by the FRU, the ambiguity in the result gives the police considerable discretion in how to classify the investigation. This issue is particularly pronounced when arrestees are taken into custody and reviewed at different times. In addition, CPD does not offer detailed guidance on how to classify cases into clearance categories, allowing officers substantial discretion in departing from a shared standard, even when a single arrestee is presented to the FRU. Lastly, if a case is rejected, the police can still file misdemeanor charges against the suspect, which, however, entail much smaller penalties, typically different from a prison sentence. If the police still file misdemeanor charges, the officers might consider the case a successful clearance by arrest and prosecution. The outcome of the felony review is, therefore, the most accurate measurement of the success of an investigation for the k-th arrestee couple. Furthermore, this maps more straightforwardly to the court outcomes. The result of the felony review and the status of the investigations are still strongly related: the correlation between acceptance and clearance by arrest and prosecution is 0.4627 in the sample used for the outcome test, which rises to 0.4915 when restricting to crimes linked to only one arrestee.[38]

5 Results

5.1 Main Results

Table 4 presents the results from the estimation of Equation 4. Column 1 shows a -9.6 pp (-3.7 pp) unadjusted gap in acceptance rates between Black (Hispanic) and White victims of violent crime, which shrinks to -7.8 pp (-1.9 pp) when accounting for the different types of crime they experience (Column 2). When accounting for the area-time fixed effects and the identity of detective at review (Column 4), arrestees who commit a violent crime are 6.6 pp (2.2 pp) less likely to enter the court process if the victim is Black (Hispanic) rather than White. This corresponds to a 7.25% gap for Black victims and a 2.2% gap for Hispanic victims.

The model upon which the test is based assumes that characteristics other than race can affect the expected rate of success of the case, and hence how fast evidence accrues and reaches the desired level, but they do not enter the cost function that pins down the required thresholds for submission. We can relax this assumption and test how the gap in success rates changes when we account for other observable characteristics of the victim, or the incident, that might correlate with race – and success. Racial bias could in fact originate *indirectly* from another characteristic that correlates with race and success (Bohren et al., 2022). In that case, by controlling for these characteristics, the estimates would shrink towards zero. However, we would still be in the presence of racial bias, despite it not *directly* originating from the race of the victim.

Table 5 performs this exercise, exploring how the baseline estimates, which are reported in Column 1, change once we progressively factor in other characteristics of the victim and the incident. Controlling for the sex and age of the victim leaves the estimates unaffected (Column 2), while including fixed effects for the quartiles of share of Black and Hispanic residents, separately, and the log of the median income in the tract decreases the differences for Black and Hispanic victims to -6 pp and -1.6 pp, respectively (Column 3). Interestingly, the coefficients relative to the racial makeup of the neighborhood are not significant, suggesting that racial bias originates from the identity of the single victim. On the other hand, a 1% increase in median per capita income in the tract is associated with a 1.5 pp increase in the probability of approval of the case, which is consistent with

[38] Table A.3 reports summary statistics for the sample used for the outcome test.

income discrimination as well, separately from racial bias. When replacing these neighborhood-level variables with census tracts fixed effects, the coefficients only marginally change (Column 4). The same is true when further controlling for the difficulty of the investigation (Column 5).[39]

Another characteristic that is not the race of the victim but might correlate with it and with the success rates of different cases is the identity of the arrestee. Since violent crime tends to be committed within-race, for the race of the arrestee to explain the results and make the true gaps null, the CCSAO and/or the police would have to be particularly lenient towards minority arrestees, which seems unlikely given prior work on policing and prosecution behavior (Rehavi and Starr, 2014; Goncalves and Mello, 2021). Empirically, I control for race, gender and age of the arrestees and the coefficients remain virtually unchanged (Column 6). While the coefficient is not highly significant, I find suggestive evidence of over-punitiveness towards Black arrestees, similarly to what other papers have documented in the US justice system (Rehavi and Starr, 2014; Raphael and Rozo, 2019).[40]

After including all the controls listed, the coefficients are -5.8 pp and -1.6 pp for Black and Hispanic victims, respectively. Note how these estimates have not dramatically changed with respect to Column 2 of Table 4, which simply accounted for the different types of crime these victims are exposed to, and even less so from those in Column 4 of the same Table, which also accounted for the identity of the detective and the area responsible for the investigation but nothing else. This is consistent with the assumption that focusing on the intensive margin, i.e. how well the case is solved, as opposed to the extensive margin, i.e. whether the case is solved, limits the confounding role of characteristics other than the victim's race and facilitates the identification of the agent's preferences.

5.2 Heterogeneity

I explore whether the estimates vary by the specific type of violent crime by estimating a heterogeneous version of the main model, where I interact the race of victim and the arrestee with dummies

[39] Note that this is exercise is not performed to compare crimes that are as observationally similar as possible, but to test whether bias arises indirectly via other characteristics that correlate with race of the victim (Hull, 2021, Bohren et al., 2022).

[40] Appendix B.3.1 discusses the nature of racial bias against minority suspects and how this can coexist with racial bias against minority victims. In short, detectives are biased against minority suspects if they perceive a relatively higher cost of a type I error if the suspect is non-White. This would show up empirically as a higher probability of acceptance among minority suspects. The overall stopping point depends on the weights placed by the detective on victims and suspects.

for the four types of crime: aggravated assaults/batteries, sexual assaults, homicides and robberies. Figure 3 shows that the coefficients are relatively constant for Black victims with all but the one for aggravated assault/batteries being around -5 pp. The latter category is the one where Black victims experience the largest amount of bias (-8.9 pp). It is striking how for both crimes where the victims is alive and playing an active role in the investigation - robberies and sexual assault - and crimes where they cannot do so - homicides, the estimates are close in magnitude. For Hispanic victims, the picture is more mixed, and they are discriminated against mostly if victims of violent crimes that have no sexual or financial motive, i.e. homicides and aggravated assault/batteries. Figure A.3 presents the estimates for the arrestees, obtained from the same regression. The main result that stands out is a remarkable tendency for over-punishment of Black and Hispanic individuals arrested in connection to a sexual assault and of Black suspects arrested for aggravated battery or assault.

I also assess how bias against Black and Hispanic victims varies with the racial composition of the neighborhood. Specifically, I estimate a heterogeneous version of the main model by interacting the race of the victim and the race of the arrestee with quartiles of the share of minority residents in the Census tract where the crime occurred. I find that the estimates for Black victims are largely similar regardless the racial makeup of the neighborhood, while those for Hispanic victims increase in magnitude when White residents are the minority in the neighborhood, but the differences are not significant (Figure 4). Interestingly, I document a remarkably large tendency for over-punishment of Black arrestees in predominantly White neighborhoods (A.4). Estimates for Hispanic arrestees display a qualitatively similar but more nuanced pattern.

I then test whether minority victims belonging to categories that might be perceived as particularly vulnerable receive a more favorable treatment than other minority victims. In particular, I test whether Black and Hispanic victims who are female, underage, or senior experience a higher probability of acceptance, reflecting lower bias, than minority adult males. I do so by estimating a regression that interacts indicators for Black and Hispanic victims with indicators for these three categories, including the main effects to account for baseline differences in those categories (as I have been doing since Column 2 of Table 5). As shown in Figure A.5, Black victims belonging to either of these three groups experience less bias than adult Black males, especially if the victim is underage. For Hispanic victims, only being under 18 ensures improved treatment. Overall, the detectives seem to discriminate mostly against minority adult males and to improve the quality of their work - and therefore the probability of acceptance of a case - if the victim belongs to a relatively more vulnerable group.

Lastly, I allow racial bias against minority victims to vary depending on the race of the arrestee. Although controlling for the race of the arrestee does not affect the racial disparities against Black and Hispanic victims (Column 6 in Table 5), this average estimate might mask important heterogeneity. Figure 5 shows that this is the case, by testing the difference in the probability of acceptance for each dyad of victim-arrestee race, relative to cases when both the victim and the arrestee are White. We can see that cases involving Black victims are significantly less likely to be accepted, relative to the omitted group, when the arrestee is also Black (-3.6 pp), and even more so if the arrestee is White (-7.1 pp). The remaining dyads do not differ significantly from cases involving White victims and arrestees, with the notable exception of cases involving White victims and Black arrestees (+2.8 pp). Chicago detectives seem to be submit relatively lower quality cases when a White offender allegedly committed a crime against a Black victim. In addition, Black arrestees' outcomes differ widely if they offend against White or Black victims, by 6.4 pp. These results are in line with previous work among criminologists (Roberts and Lyons, 2009, Vaughn, 2020) and those in Alesina and La Ferrara (2014). This evidence is also consistent with the theory of simultaneous over-policing (when suspects) and under-policing (when victims) of Black civilians (Goff, 2021, Boehme et al., 2022, Lewis and Usmani, 2022).

5.3 Robustness

Next, I test the robustness of the main results on two important issues. I start by addressing potential concerns about infra-marginality. I then evaluate whether the detective are equalizing outcomes other than FRU acceptance.

5.3.1 Addressing Infra-Marginality

The model presented in Section 4 assumes that the expected acceptance of the case - $p(t)$ - evolves continuously over time, and the cases are therefore submitted to FRU once this quantity equals the desired threshold of evidence. However, if $p(t)$ evolves discretely over time, some investigations might be considered concluded when their expected success has passed the desired threshold. This would mean that the thresholds are confounded by infra-marginal cases (Ayres, 2002).[41] Such violation seems particularly likely among cases that are solved very quickly. Among these, the evidence could be so overwhelming that no further follow-up is required. Figure 6 plots the relationship between FRU acceptance and days elapsed from when the crime is reported to the review of the case, separately for the three racial groups considered. Acceptance is first residualized with respect to the main design controls, i.e. area-time-crime type fixed effects and detective fixed effects.[42] The range of days to review is truncated to six months to minimize sparsity. There is a stark gap between the residualized acceptance rates of White and Black victims that persists throughout the range of days to review. The relationship is more nuanced for Hispanic victims. Nevertheless, I take several steps to test whether infra-marginality is a concern for the main estimates.

First, I tackle this issue by controlling for the time elapsed between the offense and the review by CCSAO. This exercise lies on the assumption that over-shooting is a function of the length of the investigation, and once we control for it, the comparisons would be cleaner. I do so by including a third degree polynomial in days elapsed until review. Columns 1 and 2 of Table 6 present the results using the baseline regression (Column 4 in Table 4) and the most saturated model (Column 6 of Table 5). The estimates are virtually unchanged. We can conclude that even among cases solved in the same amount time, the suspected offenders are significantly less likely to face court proceedings and any degree of accountability if the victim is Black or Hispanic, rather than White.

Next, I drop cases that are solved relatively early, increasing the cutoff from one day, to two days, to a one week, again for the baseline model and the most saturated one. The coefficients for Black victims decrease slightly when dropping cases solved the same when the offense occurred (Columns 3 and 4) or the within next day (Columns 5 and 6), but increase again when keeping only cases that took at least one week of investigations to reach submission. Infra-marginality does not seem to be a concern for cases involving Black victims. On the other hand, violent crimes experienced by Hispanic victims have the same probability of acceptance when focusing on investigations that lasted at least one week. Notice that this might reflect cases of Hispanic victims being infra-marginal in the short run, or the desired thresholds varying with the duration of the investigation. It is possible that detectives' standards become less lax over time, i.e. $c(\cdot)$ is a

[41] Notice that in my model the detectives are not going to submit cases that have not reached the threshold, which means that expected success can "overshoot" but not "undershoot".

[42] Figure A.6 uses the raw acceptance rates.

function of time t as well, in a way to close the gap between Hispanic and White victims, potentially also due to the self-selection of the detectives into keeping a case open for longer. Lastly, I find a substantial tendency for over-punitiveness towards Black and Hispanic arrestees when dropping cases solved immediately.

5.3.2 Other Outcomes

In the main model, I assume that the outcome the detective has in mind is the potential acceptance of the case by FRU. This is the first stage at the conclusion of the investigation, and therefore the least likely to be affected by other agents' bias and, most importantly, by exogenous events outside the control of the detective that might affect the relevant outcome. Nevertheless, it is possible that detectives account for these later stage decisions and events and optimize the potential guilt at the end of all court proceedings by the arrestee.

I test this by using data obtained by the Chicago Data Collaborative from the Circuit Court of Cook County on all court proceedings from 1984 to 2019. I match this information with the sample used so far for the outcome test using central booking numbers and restrict the sample further to arrests reviewed no later than 2016. This is done to allow sufficient time for court proceedings to ensue. In fact, if the length of court proceedings correlates both with race and the outcome, the estimates might be biased. The main outcome is whether the arrestee is associated with any guilty finding for that specific arrests. If the case never went beyond FRU, the outcome is by default a zero. Since this outcome is mechanically correlated with the number of charges brought against the individual arrestee, I further control for this variable in my models.

Table 7 reports the results. Even after controlling for all the available features of victim, arrestee and crime, suspects are 6.7 pp less likely to be found guilty if they commit violence against Black individuals than White individuals. The difference is sizeable, as it is 8.8% of the White victim mean. The magnitude of the coefficient for Hispanic victims matches that in Table 5 but it is not significant at conventional levels. Lastly, there is no evidence of significant over-punitiveness towards minority suspects in court, at least in the full sample.

Tables A.4 and A.5 consider alternative outcomes obtained from the court data. The former uses as outcome whether the arrestee is found guilty of the top charge brought against them, which is typically the one corresponding to the i-th crime. The latter takes a step back in the court process and instead adopts as outcome whether the arrestee appears in court dataset in the first, which is roughly equivalent to the arrestee appearing at arraignment. This occurs if the case passes FRU and is considered viable by the arraignment prosecutor.[43] An arrestee is 6.7 pp (2.3 pp) less likely to appear at arraignment and 8.1 pp (2.2 pp) less likely to be found guilty of the top charge brought against them if they commit violence against a Black (Hispanic) victim rather than a White victim, with all these coefficients being significant.

The objective of police officers can also be framed as crime minimization. Detectives may therefore aim to equalize, across victim race, the probability that a suspect reoffends. The model in Section 4 can be reinterpreted with $p(t)$ representing the probability that the suspect k, who is potentially unknown at the beginning of the investigation, recidivates. Under this interpretation, $p(t)$ is now non-increasing in t: as more time is invested in the investigation, the likelihood of uncovering the culprit and apprehending them increases. Bias can now be understood as a higher tolerance for recidivism, i.e. a higher threshold applied to the discriminated group. A detective biased against group r perceives future crimes by suspects who initially offended against that group as less costly. These future crimes may be committed against the same victim, which is especially likely for batteries, sexual assault and domestic violence, or at least within the same demographic group, as prior research suggests (Becker, 2007).

I perform the same test using recidivism within one or three years of the arrest of suspect k in connection with the crime against victim i. Recidivism is defined based on arrests for either *i)* violent index crimes, or *ii)* violent index crimes involving a felony. These are serious offenses that impose higher costs on victims and are less subject to under-reporting. Table A.6 reports the results. Suspects arrested for violent crimes against Black victims are 4.9 pp (2.9 pp) more likely to commit another serious violent crime within 3 years (1 year) than those arrested for violent crimes against White victims, even after controlling for differences in recidivism risk by suspect race. These effects are substantial, amounting to 16.8% (18.5%) of the White victim mean. When considering recidivism based violent crimes involving that *also* involve a felony, the differences slightly fall to 3.7 pp (13.5%) and 2 pp (12.7%) for the three- and one-year windows, respectively. For Hispanic victims, the recidivism gap is 2.9 pp and 2.5 pp over three years, both statistically significant, but shrinks and becomes not significant over a one-year horizon.

[43] In Cook County, the ASA conducting the felony review is different form the ASA taking over the case after the initial review.

Another outcome that can be used to perform the test is the status of the investigation, rather than the decision of the FRU or the court rulings. As discussed in Section 4, the status of the investigation can differ significantly from the FRU decision, especially when more than one arrestee is involved, and even more so if they are arrested and their cases reviewed at different times. It is useful to compare the results obtained using the FRU decision with those obtained with the status of the investigation, as the latter is the only information that is typically shared by police departments, when systematically collected. Table A.7 presents the results of the test using as an outcome an indicator that takes value 1 if the status of the investigation is clearance by arrest and prosecution, using the sample used so far in Column 2 and restricting to cases with only one arrestee involved in Column 4. For reference, results using the FRU decision as outcome are reported in Columns 1 and 3, respectively. The coefficients are roughly half the size when using the status of the investigation, which suggests that racial bias disparities would be under-estimated if the actual FRU decision were not available. While we cannot assess whether this is caused by intentional misclassifications or mistakes that correlate with race, these results call for improved data collection by police departments. The FBI has imposed police departments to use the NIBRS system, which entails a more detail data collection on the outcome of crime investigations, but this might still not be sufficient to have a clear picture of how many arrest enter the court process, as shown here.

5.4 Other Concerns

Table A.8 addresses other concerns. Columns 1 and 2 restrict the sample to cases involving only one victim or one arrestee, respectively. Column 3 combines these two filters. Column 4 does not weigh each observation by the inverse product of the number of victims and arrestees involved. Column 5 clusters the standard errors at the detective level rather than the area-time-crime type level. Column 6 uses the identity of the first detective assigned to the case rather than the one who is on the case at the time of the review. The results are qualitatively unaffected by these tests.

6 Mechanisms

So far, I showed that cases involving violent crimes are less likely to be accepted by the prosecution, and the be linked to a guilty ruling in court, if the victim is Black, or Hispanic, rather than White.

This is consistent with the detectives being more tolerant of a failure, i.e. lack of justice, is the victim is not White. Next, I dig deeper into the interpretation of these estimates and the drivers of this bias. First, the results so far have been framed as detectives' bias, but it is possible that the detectives hold the same standards for all victims to and CCSAO is disproportionately more demanding for cases involving Black victims. This would still be consistent with the detectives being more tolerant of a rejection for cases of Black victims, and therefore still display bias, but they would not be *directly* setting differential standards. Second, assuming that detectives play an active role, their bias can originate from two sources: biased *preferences* against minority victims (racial animus) (Becker, 1957) or biased *beliefs* regarding what the prosecution seeks (inaccurate statistical discrimination) (Bohren et al., 2023).

6.1 The Role of Detectives as Drivers of Bias

6.1.1 Variation Over Time

During the time window under analysis, 2006-2023, three different State Attorneys (SAs) have led the office: Richard Devine, a White man, at his third term, until December 2008; Anita Alvarez, a Hispanic woman, for two terms until December 2016; Kim Foxx, a Black woman, for two terms until December 2024. Besides demographic differences, the first two are more orthodox police-friendly, though-on-crime prosecutors. Kim Foxx, on the other hand, campaigned on a reformist platform that championed de-prosecution and a rehabilitative approach, which made her often clash the police department (Conroy, 2001, Spielman, 2015, Grimm, 2022). If the estimates remain consistently negative throughout the time window, and especially under the more progressive Foxx, we could conclude that the police preferences, at least to some extent, actively drive investigative behavior and, consequently, the differential rejection rates.

I estimate a heterogeneous version of the main model, where I interact race with dummies for the specific SA who was leasing the office when case i was reviewed.[44] As shown in Figure 7, the estimates are negative and significant for all terms among Black victims. Although we can reject that they are equal (the p-value for the null that they are all equal is 0.0123), we can safely reject that the gaps in rejection rates under Alvarez and Foxx are the same between Black and White victims (p-value=0.7507). The same holds for Hispanic victims, where all estimates are statistically

[44]Notice that the only included term of Richard Devine is not complete, as it misses the first year, as well as Foxx's second term, which is missing the last year.

equal (p-value=0.6897), even though none of them are individually significant . Figure A.7 presents the analogous estimates for Black and Hispanic arrestees, obtained from the same regression. No coefficient is significant at the 5% level and the magnitudes are relatively small, with the exception of Foxx second term, which is marked by a moderate tendency for over-punishment of Black arrestees.

6.1.2 Variation Across Areas

Detective areas differ widely in the management, leadership and practices. Additional evidence in favor of the police preferences playing an active role in the differential rejection rates would be the estimates varying by detective area. I therefore estimate a heterogeneous version of the main model where race of the victim and race of the arrestee are interacted with fixed effects for the detective area where the crime was investigated. Since the boundaries, and the number, of the areas changed over time, I estimate two separate regressions: one for the period from 1/2006 to 2/2012, when the city was divided into 5 areas, and one for the period 3/2012 to 3/2020 when the 5 areas were consolidated into 3. CPD went back to 5 areas in 2020, but the number of observations for this third period is too low to perform the same exercise. Figure 8 presents the results. In the first period, the estimates are relatively homogeneous, with the exception of Area 1, which shows consistently lower bias against Black and Hispanic victims. The estimates display large variation in the second period, with Area 2 standing out as the most biased one. Area 1, on the other hand, is still the unit associated with the least bias, despite the change in boundaries. Figure A.8 explores the variation in bias against Black and Hispanic arrestees. In the first period, Area 2 and Area 5 have a noticeable tendency for over-charging of Black and Hispanic arrestees. In the second period Area 2's employees display joint bias against Black and Hispanic arrestees. Given that all areas interact with the same State Attorney's office, with cases randomly assigned to ASAs, the substantial heterogeneity in the estimates across areas points towards the preferences of the police, here defined as jointly set at the area level, actively determining the differential acceptance rates by race of the victim, and hence racial bias.

6.1.3 Detective-level Estimates

If the detective thresholds matter for the acceptance of the case, and some of them systematically discriminate against minority victims, we should observe a considerable amount of between detective variation in acceptance gaps by victim race. Because cases are further randomly assigned to ASAs for felony review, the expected acceptance threshold applied by the ASA is independent of the detective assigned to the case. Consequently, variation in racial gaps in case acceptance across detectives reflects differences in detective case preparation and submission behavior rather than ASA-side bias.

For these reasons, I estimate detective-level racial bias by taking the most saturated model (Column 6 in Table 5), and interact the race of the victim with detective fixed effects, δ_j:

$$Y_{ik} = \alpha + \beta_j^B BlackVictim_i + \beta_j^H HispanicVictim_i + \gamma' X_{ik} + \delta_j + \delta_{ayc} + \varepsilon_{ik} \quad (5)$$

$\hat{\beta}_j^B$ and $\hat{\beta}_j^H$ are estimates of racial bias against Black and Hispanic victims for each individual in the sample. The ultimate goal is to estimate the variation (standard deviation) in bias.[45] These estimates are informative insofar as detectives close a sufficient number of cases for victims of all three demographic groups considered here. Therefore, I restrict the sample to detectives who close at least 5 cases for each group. Since many detectives solve only a handful of cases – either because they are assigned few cases or because they are not able to solve more – and the racial distribution of the solved cases is often skewed, I am left with only 52% of the full sample and 17% of the detectives in the full sample (327 out of 1,885).[46]

Taking the variance of the raw detective-level estimates ($\hat{\beta}_j^r$) would overstate the true variation in bias due to sampling error (Walters, 2024). To account for this, I use an empirical Bayes approach. I assume a standard normal–normal hierarchical model for the true detective-level bias (β_j^r) and the sampling distribution of the individual estimates ($\hat{\beta}_j^r$) (Morris, 1983). In this framework, I first construct a detective-specific, design-based estimate of the sampling variance of each $\hat{\beta}_j^r$, and then estimate the true between-detective variance by subtracting the average sampling noise from the observed variance of the raw estimates.[47]

I find a substantial amount of variation in bias: the standard deviations in bias against Black

[45]The average detective-level bias is obviously also of interest, but we have already obtained a first estimate of this parameter in the main results.

[46]Another undesirable consequence of this sample restriction is that most of the sample is represented by robberies, namely 69%, while they constitute less than 52% of the full sample used for the outcome test. This is due to the fact that robbery detectives are those who tend to work on, and therefore solve in absolute terms, disproportionately more cases.

[47]Notice that this is not the variance of the shrunken estimates, which instead would understate the true variation in bias. See Appendix B.3.2 for details.

victims and bias against Hispanic victims are 13.8 pp (SE=0.8 pp) and 12.2 pp (SE=1.9 pp), respectively. These are sizeable quantities. The standard deviation in bias against Black victims amounts to 16.4% of the raw acceptance rate among cases involving Black victims (84%) and 170% of the unadjusted Black victims-White victim gap (-8.1%). Similarly, the standard deviation in bias against Hispanic victims represents 13.7% of the raw acceptance rate among cases involving Hispanic victims (89%), and it is 3.5 times larger than the unadjusted Black victims-White victim gap (-2.7%). This considerable amount of heterogeneity in the detectives' acceptance gaps suggests that their attitudes matter, especially given that reviews are randomly assigned to ASAs.[48]

6.2 Sources of Racial Bias: Leveraging the Detective's Characteristics

The literature on discrimination has pointed out that what models of discrimination generally attribute to biased preferences against minority individuals (racial animus) could instead originate from biased beliefs (inaccurate statistical discrimination). Here, these would be biased beliefs held by the detectives about what makes an investigation successful upon completion (Bohren et al., 2023). It is in fact difficult to distinguish between the two, and both correspond to forms of bias (Arnold et al., 2018; Arnold et al., 2022). For policy reasons, however, it is important to understand whether we should intervene on beliefs and information to correct racial inequities or if some other levers have to be pulled, such as de-biasing training. It is important to note that racial animus in this setting may stem from several different sources, which cause $c(m)$ to be lower than $c(w)$. $c(rv)$, in fact, includes personal racial attitudes of the detective, the organizational incentives from concluding a successful investigation for a victim of a given race, the perceptions of the public, and potentially more. I attempt to distinguish between these two main forms of bias - biased beliefs and biased preferences - by leveraging the characteristics of the detectives who clear the case, similarly to what Arnold et al. (2018) do with bail judges.

If inaccurate statistical discrimination explains the main results, cases of Black victims would be disproportionately more likely to be rejected by the ASAs because detectives hold inaccurate beliefs regarding evidentiary quality and/or standards rather than biased preferences. Either detectives tend to disproportionately over-estimate the quality of the evidence when the victim is non-White,

[48]Figure A.9 plots the distribution of raw and shrunk fixed effects with the estimated hyper-parameters of the prior distribution (mean and standard deviation). Table A.9 replicates the main specifications in the sample used to estimate detective-level bias. The average bias is remarkably almost the same as in the full sample.

or they under-estimate the standards required by FRU. Either way, they inaccurately map evidence into expected outcomes. If biased beliefs were driving the results, it is reasonable to assume that as a detective accumulates experience, the racial gaps would shrink, as she updates her beliefs. If we do not find any significant interaction between experience of the detective and the race of the victim, racial animus is a more likely explanation for the observed bias.

I compute four measures of experience: *i)* the number of years between the review of a case and the first date as a detective; *ii)* the cumulative number of cases assigned to detective j before the review of the case of the victim-arrestee pair ik; *iii)* the cumulative number of reviews submitted by detective j before review of the case of the victim-arrestee pair ik; *iv)* the cumulative number of rejections experienced by detective j before review of the case of the victim-arrestee pair ik. Since the case assignment data starts in 2001 and the review information begins in 2006, these measures are available for different subsets of detectives: *i)* is available for all of them, *ii)* is available for detectives appointed from 2001 onward, *iii)* and *iv)* for detectives appointed from 2006 onward. I restrict the samples accordingly in each specification. Since the potential effect of experience is reasonably nonlinear, I use a quadratic function both as baseline and in its interaction with the race of the victim.

Table 8 reports the results of these regressions. I find marginally diminishing returns to experience across all definitions of experience, with the exception of the cumulative number of cases submitted prior to a given case, for victims of all races. Experiencing a rejection appears particularly important for the success of future submissions, consistent with detectives learning from their failures and adjusting their practices to increase the probability that subsequent cases are accepted.[49] On the other hand, experience does not have any differential effect for Black or Hispanic victims, as all interactions are small in magnitude and not significant. Gaining experience on the job does not seem to close the racial gaps in acceptance rates, regardless of the specific definition of experience adopted. We can therefore reject that inaccurate beliefs on evidentiary quality and/or CCSAO's requirements are an explanation for the main results.[50] Consequently, we can conclude

[49] After a rejection, detectives may either submit the next case with stronger evidence or refrain from submitting a case they now expect would not be accepted. In both cases, they are effectively targeting a desired acceptance rate, as assumed in the model.

[50] More specifically, this requires the following assumptions: *i)* experience is a good proxy for the richness of the detective's information set; *ii)* the detective's preferences do not become more egalitarian over time in a way that would offset a simultaneous correction of her beliefs. Given that the results hold across several different definitions of experience, whose marginal increments occur over very different time spans, *ii)* seems likely to hold.

that detectives display racial animus, i.e. the main results are driven by biased preferences against minority victims among the detectives.

While we can rule out biased beliefs with relatively modest assumptions, more directly corroborating that the observed disparities are driven by biased preferences – and, even more so, pinpointing which characteristics of detectives or their environment predict such preferences – is more difficult. Previous work on discrimination by law enforcement agents has tested for biased preferences by studying whether the rankings of the outcome, by race of civilian, are a function of the race of the decision-maker (Anwar and Fang, 2006, Antonovics and Knight, 2009, West, 2018). More recent work however has pointed out that this exercise is of limited informativeness (Arnold et al., 2022). I still perform this exercise by checking if cases where the race of the detective and the victim match display higher success rates than those where they don't. Table A.10 shows no evidence of race concordance. This means that assigning a Black or Hispanic detective to a case of a Black or Hispanic victim, respectively, does not shrink the gaps in acceptance rates. This ultimately suggests that racial bias in this setting is not a function of the detectives' race. There are a few alternative explanations. Detectives might be responding to incentives introduced by other agents. For example, upper management might differentially reward, or impose, acceptance rates of victims of different races. This is related to another potential explanation, which is the preferences of the public. Crimes committed against White victims might be more salient and a failure to bring them justice could be sanctioned more harshly.

6.3 Related Bias from External Agents: Local Newspapers

Detectives might not be the only agents displaying bias against Black/Hispanic victims. Their actions might in fact be reflecting the views of the public and/or the amount of external scrutiny they face. Lower coverage of minority homicides decreases the pressure on detectives to deliver successful outcomes, which in terms of my model would translate into lower desired evidentiary standards for these victims, $c(m)$. I test this claim by collecting all the mentions of each victim of homicide – the only type of crime for which names are disclosed – in Chicago from 1/1/2001 to 12/31/2023 that appeared in the print version of the Chicago Tribune or Chicago Sun Times, the main local newspapers in Chicago, within $k \in [1, 7]$ days of the death of the victim.[51] I can then

[51] A victim might not die immediately after the incident occurred, but names are typically disclosed by CPD only upon death.

compute whether a victim is mentioned and how many articles are published. Figure 9 plots the cumulative probability of mention by either of the two newspapers in Chicago within the first 7 days after the death of the victim. We observe that if a victim is not mentioned by day 3, they are rarely mentioned thereafter. A wedge between White and Black victims is visible from the very first days. Hispanic victims are less likely to be mentioned than White victims but more likely to be mentioned than Black victims.[52]

Given that coverage plateaus after the third day and that the more time passes, the more coverage is itself influenced by policing activities, I focus on cumulative coverage by the third day – third day included. I test whether coverage differs by race of victims in the first days after they die, controlling for their other demographics (age, gender), time variables that affect whether the news can appear in the printed paper and media-related time trends (day of the week, hour of day, month, year), or the newsworthiness of the homicide (weapon, whether it is gang related, whether is is domestic violence related, neighborhood characteristics). Specifically, I estimate the following model:

$$Y_{iy} = \alpha + \beta^B BlackVictim_i + \beta^H HispanicVictim_i + \gamma' X_{iy} + \varepsilon_{iy} \tag{6}$$

where Y_{iy} is either a dummy for whether the victim i, who died at time y has been mentioned in local newspapers by the third day after the event or the total number of mentions by the third day. Given the skew in the distribution of total mentions, a Poisson model is used for this second dependent variable rather than OLS. If a crime is solved within three days, it might mechanically receive more coverage due to the higher amount of information (arrest of the suspect, charges, etc.). To account for this I control whether the crime is solved within three days, using the clearance date of the case, which is well recorded for homicides. This is obviously a potentially *bad control* (Angrist and Pischke, 2009), but the results without this additional regressor are qualitatively very similar.

I document two main facts (Table 9): *i)* Black victims, and to a lesser extent Hispanic victims, are less likely than White victims to be mentioned in the paper at all, and ii) fewer articles are written about their death. Coverage is also significantly lower in predominantly minority neigh-

[52] Note that, as this is printed media, coverage is not immediate; if a homicide occurs after the daily printing deadline, the homicide will by definition not be covered the following day. Moreover, the police might also not release information about the victim immediately, and the paper might also desire conduct some follow-up on the victim and incident before the coverage.

borhoods. Note that the White–Black disparities persist even when including tract fixed-effects, at least on the intensive margin. On the extensive margin, the within-tract difference is not significant but remains sizeable, as it represents almost 10% of the White victim mean. In addition, I find that heterogeneity in coverage bias across victim characteristics, using the total number of mentions by the third day, closely mirrors the heterogeneity in detectives' submission bias displayed in Figure A.5. Black and Hispanic homicide victims under age 18 receive much more coverage than minority adult males; victims over age 60 are second in the ranking of groups with a positive interaction, especially among Black victims; whereas being a female minority victim does not increase media coverage (Figure A.10). Although estimating the causal effect of this gap in coverage on the clearance and success gaps is beyond the scope of this paper, this set of results provides suggestive evidence that the equilibrium in the media market is one of lesser interest in non-White victims. Detectives might therefore best-responding to the amount of scrutiny they face from the media and the public. Nevertheless, they display bias against non-White victims.

7 Policy Implications and Conclusion

This paper provides the first evidence that the police discriminate by race when interacting with victims of crime. I do so by focusing on police investigations of violent crime. I develop a novel marginal outcome test for racial bias that does not require random assignment of cases to decision-makers and that exploits the dynamic nature of the setting. By leveraging the prosecution's decision to accept or reject a case based on the evidence presented as the outcome of interest, I show that the detectives in the Chicago Police Department systematically close lower-quality investigations if the victim is Black relative to when the victim is White. For both fatal and non-fatal violent crime, Black victims are in fact less likely to ever see their assailant go to court. This is true regardless of the race of the alleged perpetrator and persists when looking at other court outcomes. As detectives' experience does not seem to reduce these gaps, the estimates are unlikely driven by the detectives missing key information or having biased beliefs in general.

My results highlight several policy implications. First, calls for oversight and transparency in policing, which have been increasing in recent years, should also consider investigations, not only activities such as traffic stops and use of force, which have come under major scrutiny. While less salient in the public's mind than other policing activities when discussing policing reforms, the decisions of police investigators can also be biased, and they should be the object of external monitoring. Moreover, this paper documents that only a share of all crimes that are considered solved by the police actually end up entering the court process, due to the prosecution deeming the evidence presented by the police to be insufficient. The issue becomes even more sizable if we consider non-fatal violent crime and the remarkable share of clearances achieved due to the victim's refusal to cooperate. For these reasons, the overall clearance rate seems an inadequate metric to measure the performance of a police department, a concern echoed by Cook and Mancik (2024) and Baughman (2020).

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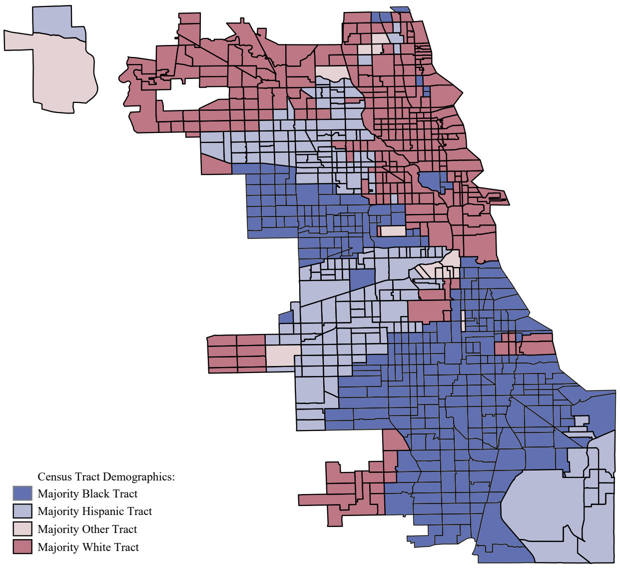
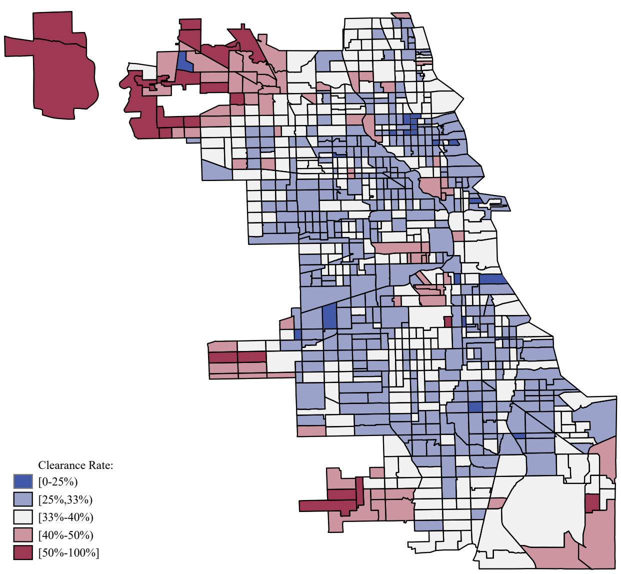
Figures and Tables

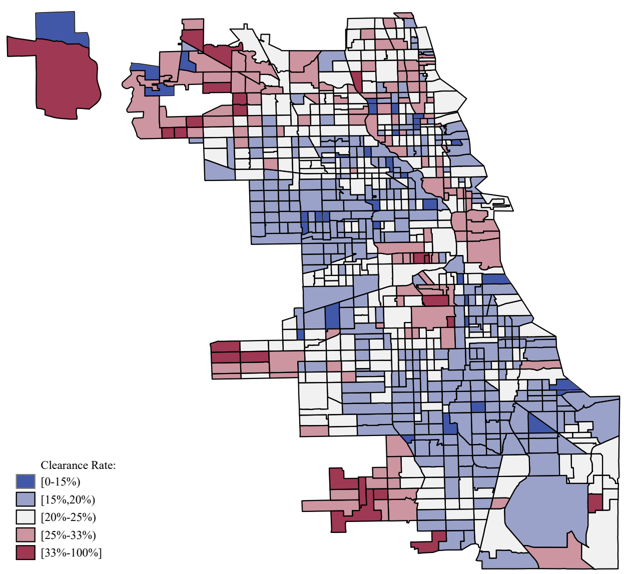
Figure 1: Racial/Ethnic Majority and Clearance Rates by Census Tract



(a) Racial/Ethnic Majority



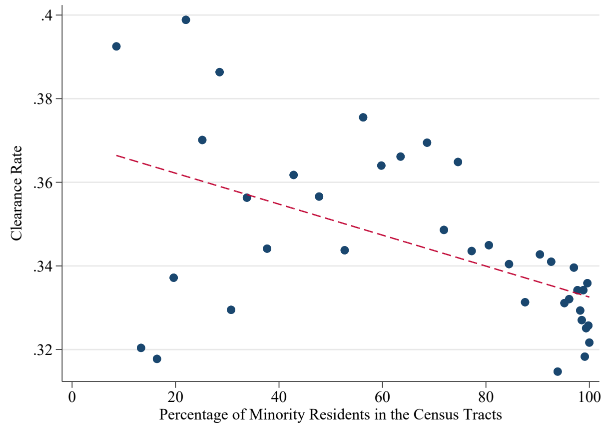
(b) Clearance Rates



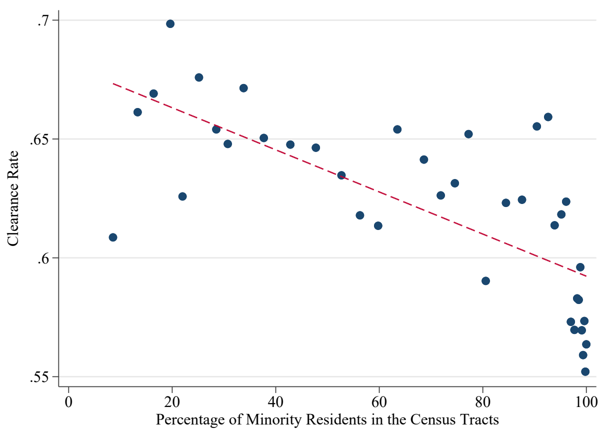
(c) Clearance Rates - Arrest and Prosecution

Notes: This Figure shows the demographic composition and the clearance rate over the time period 2001-2023 in Chicago Census tracts. Demographic information comes from the 2010 ACS.

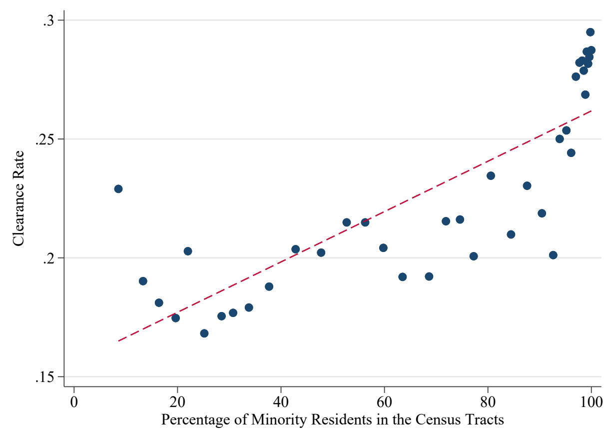
Figure 2: Relationship Between Clearance Rates and Percent Minority Residents in the Census Tract



(a) Clearance Rate



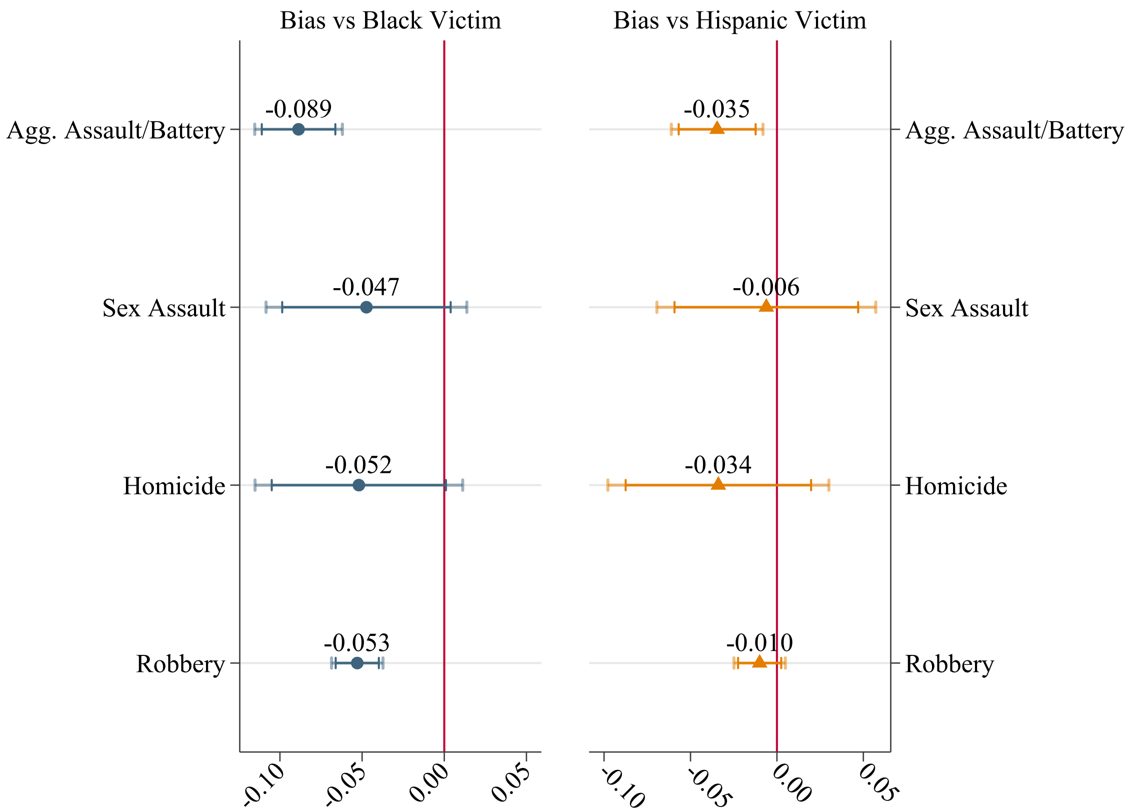
(b) Arrest and Prosecution | Clearance



(c) Victim's Refusal to Cooperate | Clearance

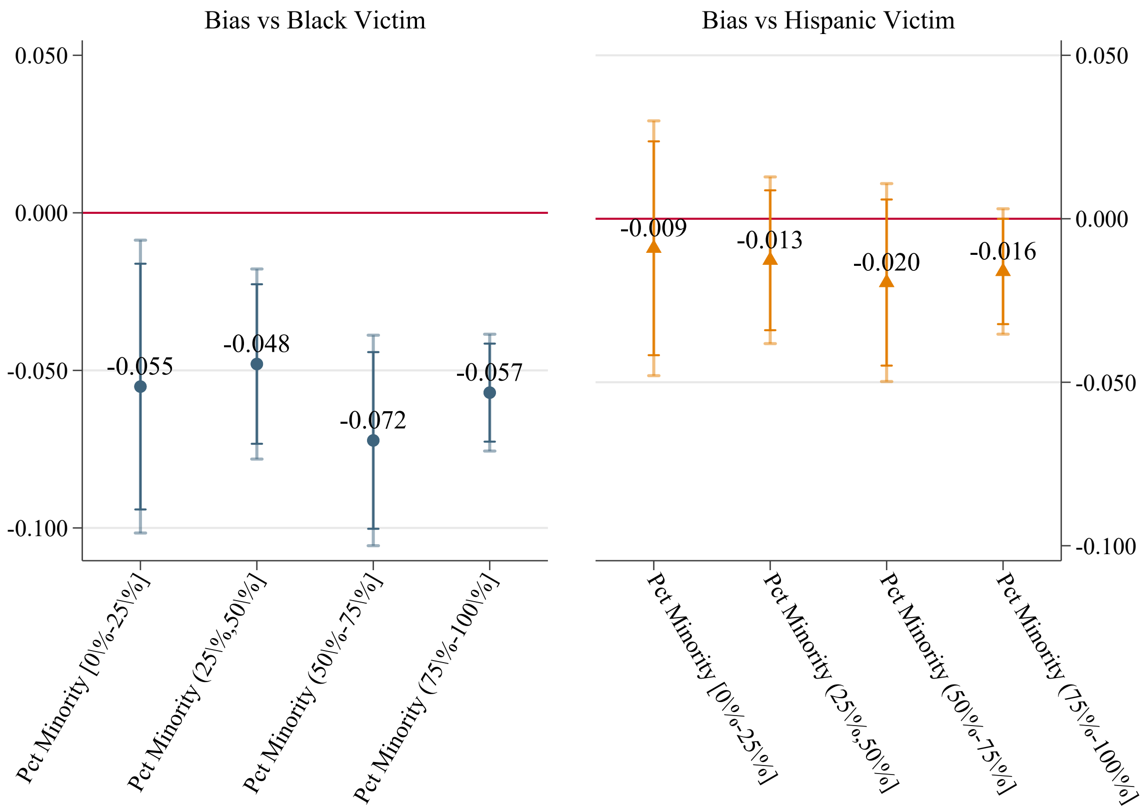
Notes: This Figures shows a binned scatterplot of the relationship between clearance rates and share of minority residents in Chicago Census tracts, with a linear fit. Panel a) uses the overall clearance rate, while panels b) and c) display the conditional clearances rates by arrest and prosecution and by exceptional means due to the victim's refusal to cooperate.

Figure 3: Heterogeneity in Racial Bias by Type of Violent Crime



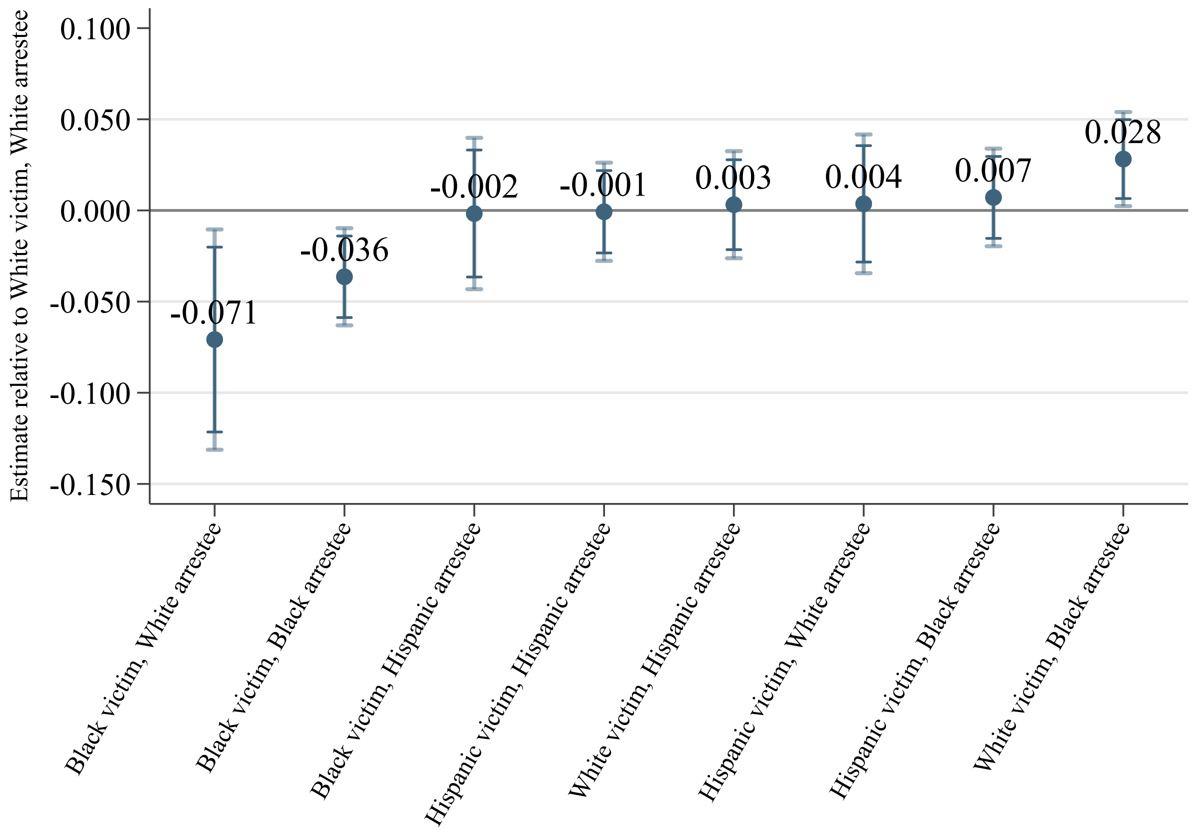
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5 by type of violent crime. Race dummies for victims and arrestees are interacted with fixed effects for the type of violent crime. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 4: Heterogeneity in Racial Bias by Racial Makeup of the Neighborhood



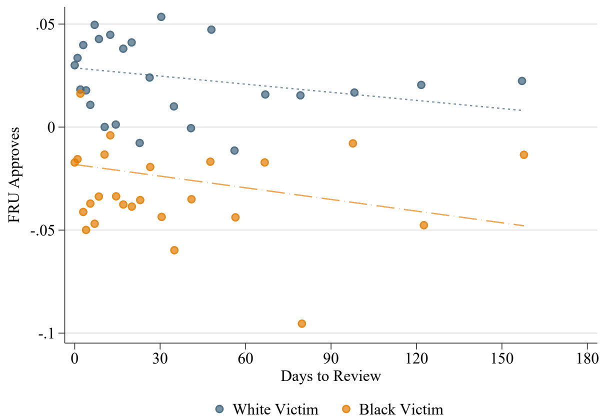
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5. Race dummies for victims and arrestees are interacted with fixed effects for the quartile of the share of minority residents in the Census tract. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 5: Heterogeneity in Racial Bias by Victim-Arrestee Dyad

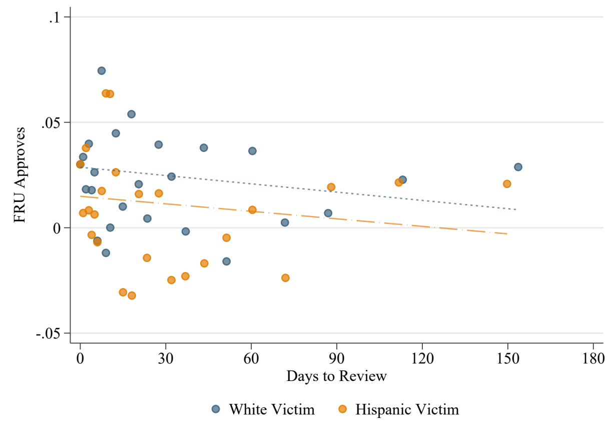


Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5. Race dummies for victims are interacted with race dummies of arrestee race. The omitted cases are those with White victims and White arrestees. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 6: Relationship between Residualized Acceptance and Days to Clearance



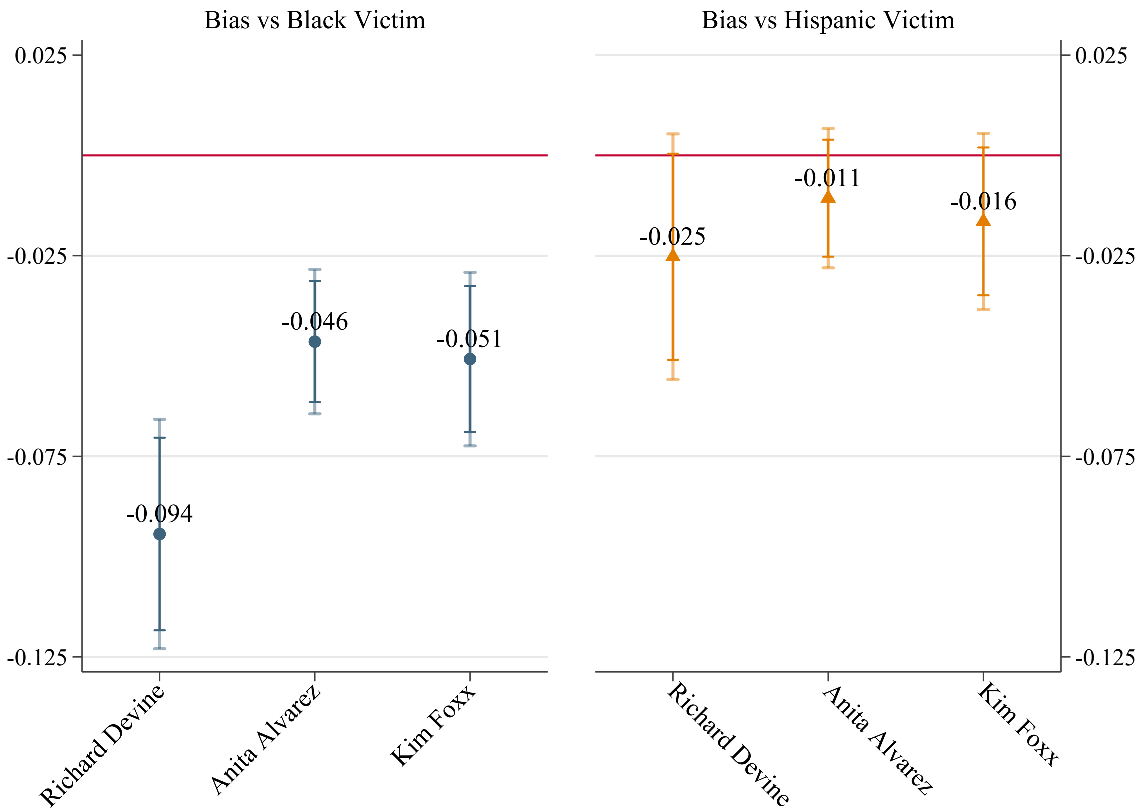
(a) White Victim vs. Black Victim



(b) White Victim vs. Hispanic Victim

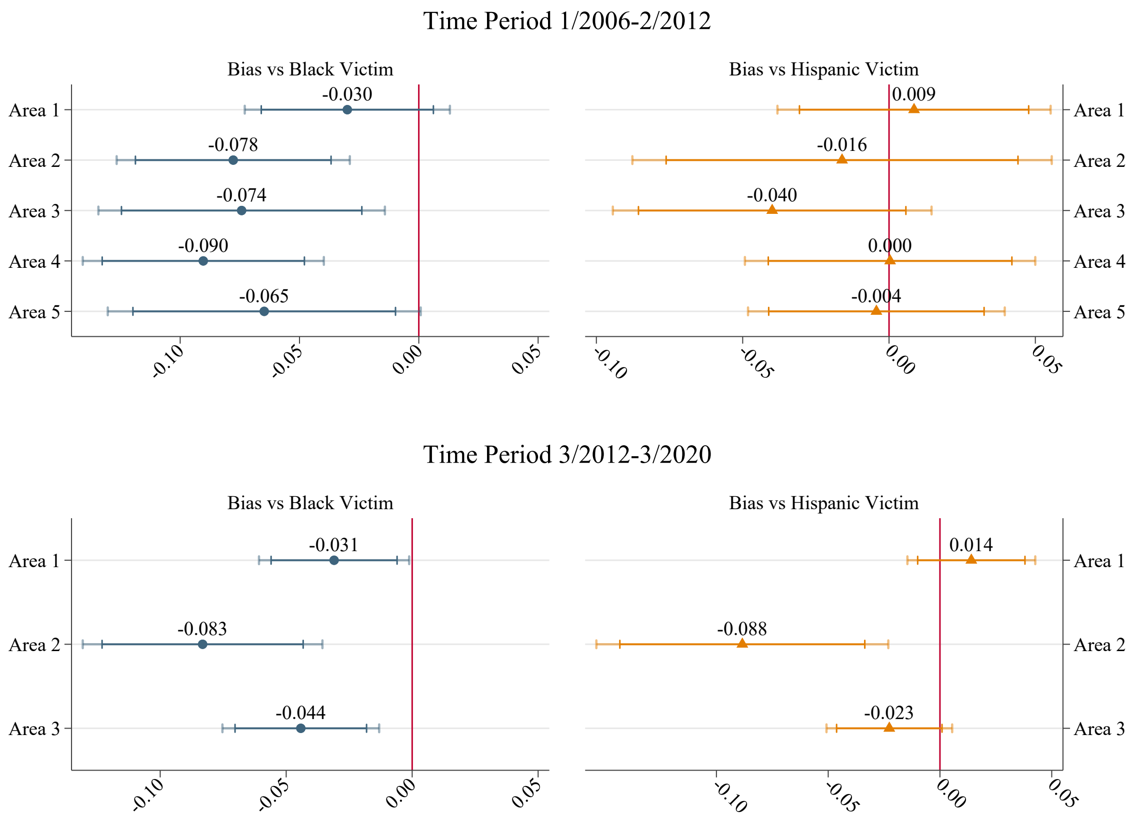
Notes: This figure plots a bin scatter of the relationship between acceptance by FRU and days elapsed from report of the crime to review, by race of the victim. Acceptance is first residualized with respect to area-time-crime type fixed effects and detective fixed effects. The range of days to review is truncated at 180 days.

Figure 7: Heterogeneity in Racial Bias by SA's Term



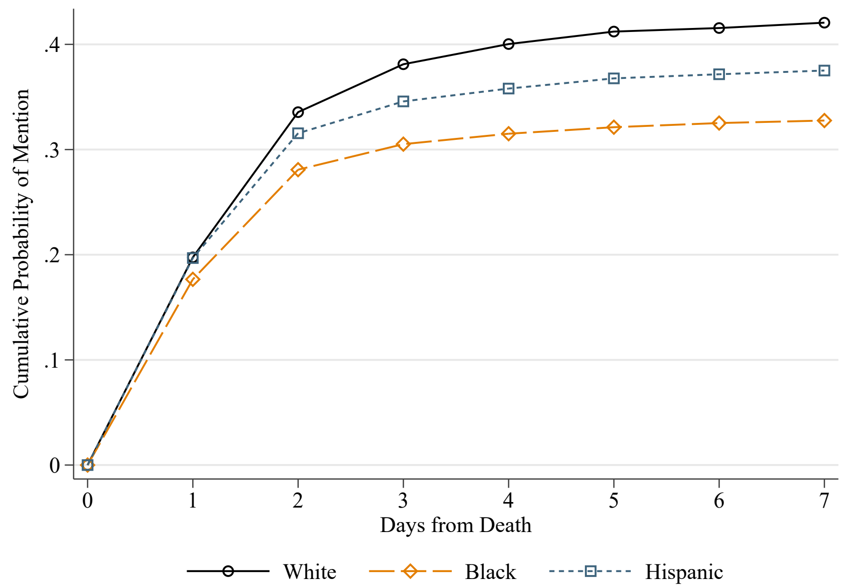
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5 by SA leading the CCSAO. Race dummies for victims and arrestees are interacted with fixed effects for the SA leading the CCSAO. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 8: Heterogeneity in Racial Bias by Detective Area



Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5. Race dummies for victims and arrestees are interacted with fixed effects for detective area where the crime occurred and investigated. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure 9: Cumulative Probability of Mention of Homicide Victims in Chicago by Race of the Victim



Notes: This figure plots the cumulative probability that a victim is mentioned in the print version of either the Chicago Tribune or the Chicago Sun Times in the first 7 days after the death of the victim. The statistics are shown separately for White, Black and Hispanic victims.

Table 1: Characteristics of the Victims, by Race

	Homicide			***Non-Fatal Violent Crime***		
	White Victims	***Black Victims***	***Hispanic Victims***	***White Victims***	***Black Victims***	***Hispanic Victims***
Incident/Victim Information:						
Female Victim	0.241	0.108	0.103	0.355	0.403	0.292
Victim's Age	39.046	29.453	26.917	35.753	32.126	30.679
Gang Related	0.123	0.144	0.383	0.018	0.022	0.098
Domestic Violence	0.137	0.062	0.055	0.075	0.219	0.118
Firearm	0.530	0.854	0.828	0.255	0.362	0.345
Assault				0.161	0.205	0.198
Battery				0.196	0.371	0.284
Sexual Assault				0.072	0.054	0.056
Robbery				0.571	0.371	0.462
Status Information:						
Cleared	0.660	0.450	0.453	0.325	0.332	0.306
Exceptional Clearance	0.156	0.139	0.115	0.115	0.154	0.110
Arrest and Prosecution	0.504	0.311	0.337	0.210	0.178	0.196
Victim Refuse to Cooperate				0.065	0.108	0.066
Detectives Information:						
Detective Assigned	0.988	0.998	0.996	0.881	0.868	0.875
Black Detective	0.062	0.123	0.067	0.081	0.211	0.076
Hispanic Detective	0.133	0.122	0.148	0.154	0.126	0.214
White Detective	0.780	0.742	0.766	0.733	0.644	0.679
Total Victims	618	9,792	2,273	102,165	477,724	154,141

Notes: This table shows the average characteristics of the victims of violent crime, with related information about the incident, investigation and assigned detective. The information is shown separately for White, Black and Hispanic victims and for homicides and nonfatal violent crime, and covers the time period 2001-2023. Only White, Black and Hispanic victims are kept in the sample.

Table 2: Characteristics of the Detectives

White Detective	0.658
Black Detective	0.157
Hispanic Detective	0.162
Native/AAPI Detective	0.023
Female Detective	0.189
Cases per Year	24.252
Share of Modal Crime Assigned	0.837
Specialist (Modal $\geq$ 75%)	0.723
Specialist (Modal $\geq$ 80%)	0.662
Specialist (Modal $\geq$ 90%)	0.494
(Mostly) Homicide Detective	0.036
(Mostly) Sex Assault Detective	0.084
(Mostly) Robbery Detective	0.304
(Mostly) Assault/Battery Detective	0.577
Detectives	3261

Notes: This table shows the average characteristics of the detectives in the sample. The dataset is at the detective level. Cases involving victims of all races are used for this analysis. The data covers the time period 2001-2023.

Table 3: Disparities in Clearance Probability

	(1) Cleared	(2) Cleared	(3) Cleared - No Victim Refusal	(4) Cleared - No Victim Refusal	(5) Cleared - Arrest and Prosecution	(6) Cleared - Arrest and Prosecution
Black Victim	-0.032***	-0.030***	-0.043***	-0.042***	-0.038***	-0.038***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Hispanic Victim	-0.027***	-0.024***	-0.020***	-0.019***	-0.013***	-0.013***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Pct Black (25%,50%]	0.000		-0.002		-0.002	
	(0.003)		(0.003)		(0.002)	
Pct Black (50%-75%]	-0.011***		-0.011***		-0.007***	
	(0.004)		(0.003)		(0.003)	
Pct Black (75%-100%]	-0.029***		-0.028***		-0.023***	
	(0.003)		(0.003)		(0.003)	
Pct Hispanic (25%,50%]	-0.008***		-0.008***		-0.005**	
	(0.003)		(0.003)		(0.002)	
Pct Hispanic (50%-75%]	-0.020***		-0.018***		-0.012***	
	(0.003)		(0.003)		(0.003)	
Pct Hispanic (75%-100%]	-0.032***		-0.025***		-0.017***	
	(0.004)		(0.003)		(0.003)	
Ln(Median Per Capita Income)	0.001		0.005***		0.005***	
	(0.002)		(0.002)		(0.001)	
Area × Time × Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
Tract FE		✓		✓		✓
White Victim Mean	0.2776	0.2776	0.2179	0.2179	0.1657	0.1657
Observations	643,997	643,997	643,997	643,997	643,997	643,997
R^2	0.1851	0.1873	0.1248	0.1270	0.1071	0.1095

Notes: This table shows the results from the estimation of Model 1. The outcome in Columns 1 and 2 is overall clearance; the outcome in Columns 3 and 4 is an indicator for a clearance not achieved due to the victim's refusal; the outcome in Columns 3 and 4 is an indicator for a clearance achieved explicitly via arrest and prosecution. Case controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Each observation is weighted by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Gap in Acceptance Rate by Race of the Victim

	(1) FRU Approves	(2) FRU Approves	(3) FRU Approves	(4) FRU Approves
Black Victim	-0.096***	-0.078***	-0.074***	-0.066***
	(0.005)	(0.005)	(0.006)	(0.006)
Hispanic Victim	-0.037***	-0.019***	-0.020***	-0.022***
	(0.006)	(0.005)	(0.006)	(0.006)
Crime Type FE		✓		
Area × Review Month-Year × Crime Type FE			✓	✓
Detective FE				✓
White Victim Mean	0.9103	0.9103	0.9103	0.9103
Observations	43,755	43,755	43,755	43,755
R^2	0.0115	0.0494	0.1856	0.2605

Notes: This table shows the results from the estimation of Equation 4. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Columns 1 presents the raw disparity between victims of different races. Column 2 adds fixed effects for the type of violent crime the victim has experienced. Column 3 interacts crime fixed effects with area-by-time fixed effects. Column 4 adds detective fixed effects. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Adding Non-Race Characteristics of the Victim and the Incident

	(1) FRU Approves	(2) FRU Approves	(3) FRU Approves	(4) FRU Approves	(5) FRU Approves	(6) FRU Approves
Black Victim	-0.066***	-0.066***	-0.060***	-0.058***	-0.058***	-0.058***
	(0.006)	(0.006)	(0.007)	(0.007)	(0.007)	(0.007)
Hispanic Victim	-0.022***	-0.021***	-0.016**	-0.017**	-0.016**	-0.016**
	(0.006)	(0.006)	(0.006)	(0.007)	(0.007)	(0.007)
Black Arrestee						0.019*
						(0.011)
Hispanic Arrestee						0.012
						(0.011)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓	✓	✓
Tract Racial Composition and Median Income			✓			
Tract FE				✓	✓	✓
Difficulty Controls					✓	✓
Arrestee Gender and Age						✓
White Victim Mean	0.9103	0.9103	0.9103	0.9103	0.9103	0.9103
Observations	43,755	43,755	43,717	43,755	43,755	43,755
R^2	0.2605	0.2615	0.2619	0.2835	0.2936	0.2968

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Column 1 presents the baseline estimate without any additional controls. Difficulty controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Addressing Potential Infra-Marginality

	Controlling for Days		Days $\geq$ 1		Days $\geq$ 2		Days $\geq$ 7	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.066***	-0.058***	-0.062***	-0.053***	-0.058***	-0.048***	-0.062***	-0.055***
	(0.006)	(0.007)	(0.007)	(0.008)	(0.009)	(0.010)	(0.011)	(0.013)
Hispanic Victim	-0.022***	-0.017**	-0.023***	-0.020***	-0.021**	-0.015	-0.020*	-0.011
	(0.006)	(0.007)	(0.007)	(0.008)	(0.008)	(0.010)	(0.011)	(0.012)
Days to Review	-0.000***	-0.000***						
	(0.000)	(0.000)						
Days to Review^2	0.000**	0.000**						
	(0.000)	(0.000)						
Days to Review^3	-0.000**	-0.000**						
	(0.000)	(0.000)						
Black Arrestee		0.021*		0.029**		0.047***		0.035*
		(0.011)		(0.012)		(0.016)		(0.020)
Hispanic Arrestee		0.014		0.020		0.039**		0.021
		(0.011)		(0.013)		(0.017)		(0.021)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓	✓	✓
Victim Gender and Age		✓		✓		✓		✓
Arrestee Gender and Age		✓		✓		✓		✓
Tract FE		✓		✓		✓		✓
Difficulty Controls		✓		✓		✓		✓
Observations	43,231	43,231	35,381	35,376	23,800	23,788	16,410	16,393
R^2	0.2614	0.2982	0.2930	0.3341	0.3680	0.4203	0.4461	0.5115

Notes: This table shows the results from the estimation of Equation 4 with additional controls and sample restrictions. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Guilt as Outcome

	(1) Guilty	(2) Guilty	(3) Guilty	(4) Guilty
Black Victim	-0.087***	-0.070***	-0.069***	-0.067***
	(0.009)	(0.011)	(0.011)	(0.011)
Hispanic Victim	-0.030***	-0.018*	-0.017	-0.017
	(0.010)	(0.010)	(0.011)	(0.011)
Number of Charges	0.023***	0.023***	0.023***	0.023***
	(0.003)	(0.003)	(0.003)	(0.003)
Pct Black (25%,50%]		-0.005		
		(0.015)		
Pct Black (50%-75%]		-0.014		
		(0.019)		
Pct Black (75%-100%]		-0.035*		
		(0.018)		
Pct Hispanic (25%,50%]		-0.006		
		(0.015)		
Pct Hispanic (50%-75%]		-0.013		
		(0.017)		
Pct Hispanic (75%-100%]		-0.037*		
		(0.020)		
Ln(Median Per Capita Income)		0.005		
		(0.011)		
Black Arrestee				0.016
				(0.016)
Hispanic Arrestee				0.009
				(0.017)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.7632	0.7632	0.7632	0.7632
Observations	30,899	30,881	30,894	30,894
R^2	0.2468	0.2560	0.2875	0.2899

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is linked with any guilty ruling in court. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Heterogeneity by Detective's Experience at Clearance

	All Detectives	Appointed in 2001+	Appointed in 2006+	
	Years as Detective	Prior Cases (Std)	Prior Reviews	Prior Rejections
	(1)	(2)	(3)	(4)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.064***	-0.055***	-0.044***	-0.036***
	(0.014)	(0.008)	(0.015)	(0.014)
Hispanic Victim	-0.005	-0.019**	0.004	0.010
	(0.014)	(0.008)	(0.016)	(0.015)
Experience	0.018***	0.042***	0.002**	0.053***
	(0.006)	(0.011)	(0.001)	(0.006)
Experience^2	-0.000***	-0.004*	-0.000	-0.001***
	(0.000)	(0.002)	(0.000)	(0.000)
Black Victim × Experience	0.001	-0.001	-0.000	-0.006
	(0.003)	(0.010)	(0.001)	(0.005)
Black Victim × Experience^2	-0.000	-0.002	0.000	0.000
	(0.000)	(0.003)	(0.000)	(0.000)
Hispanic Victim × Experience	-0.005	-0.001	-0.000	-0.005
	(0.004)	(0.010)	(0.001)	(0.006)
Hispanic Victim × Experience^2	0.000	0.000	0.000	0.000
	(0.000)	(0.003)	(0.000)	(0.000)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Defendant Race, Gender and Age	✓	✓	✓	✓
Observations	42,746	32,544	20,295	20,295
R^2	0.2977	0.3251	0.3683	0.3824

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest in Columns 1 and 2 are the interaction between dummies for the race of the victim and the experience of the detective at the time of review, and the baseline coefficient for experience. In Columns 3 and 4, the main regressor of interest is a dummy that takes value 1 if the race of the victim and the race of the detective match. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

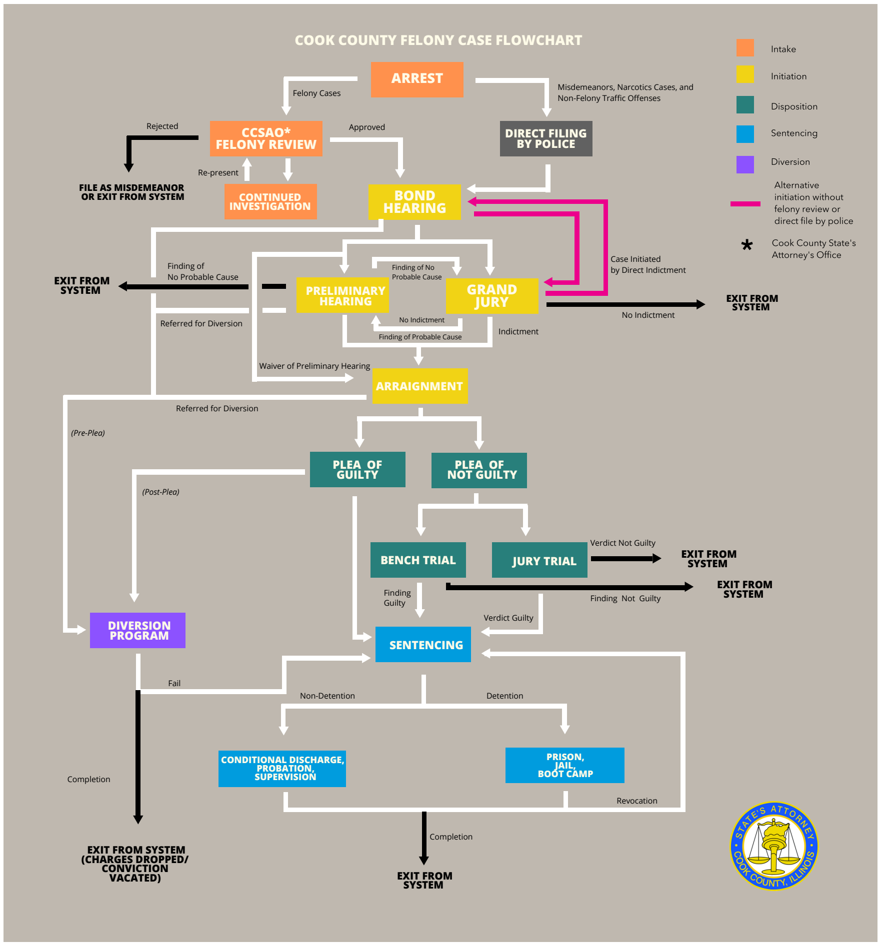
Table 9: Differential Media Coverage by Race of the Victim

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) Poisson	(6) Poisson	(7) Poisson	(8) Poisson
	Any Mention by Day 3	Any Mention by Day 3	Any Mention by Day 3	Any Mention by Day 3	Mentions by Day 3	Mentions by Day 3	Mentions by Day 3	Mentions by Day 3
Black Victim	-0.057***	-0.060***	-0.036*	-0.031	-0.369***	-0.371***	-0.205**	-0.194*
	(0.018)	(0.018)	(0.020)	(0.021)	(0.087)	(0.087)	(0.099)	(0.104)
Hispanic Victim	-0.022	-0.044**	-0.025	-0.020	-0.245***	-0.316***	-0.192**	-0.150
	(0.020)	(0.022)	(0.024)	(0.026)	(0.088)	(0.089)	(0.095)	(0.093)
Pct Black (25%,50%]			-0.008				-0.122	
			(0.022)				(0.100)	
Pct Black (50%-75%]			-0.053				-0.147	
			(0.032)				(0.105)	
Pct Black (75%-100%]			-0.057**				-0.384***	
			(0.025)				(0.125)	
Pct Hispanic (25%,50%]			-0.034				-0.342***	
			(0.024)				(0.097)	
Pct Hispanic (50%-75%]			-0.077***				-0.340***	
			(0.026)				(0.119)	
Pct Hispanic (75%-100%]			-0.045				-0.346***	
			(0.032)				(0.128)	
Ln(Median Per Capita Income)			0.024*				0.102	
			(0.014)				(0.072)	
Solved by Day 3	✓	✓	✓	✓	✓	✓	✓	✓
Age, Gender		✓	✓	✓		✓	✓	✓
Weapon, Gang, DV FE		✓	✓	✓		✓	✓	✓
Offense Hour, DOW FE		✓	✓	✓		✓	✓	✓
Offense Month, Year FE		✓	✓	✓		✓	✓	✓
Tract FE				✓				✓
% β^B					-30.850***	-30.976***	18.530**	17.649**
					(6.045)	(6.021)	(8.065)	(8.599)
% β^H					-21.760***	-27.060***	-17.481**	-13.922*
					(6.924)	(6.488)	(7.799)	(8.035)
White Outcome Mean	0.393	0.394	0.393	0.388	0.924	0.927	0.927	0.989
Observations	12,682	12,670	12,653	12,604	12,682	12,670	12,653	12,404

Notes: This table shows the results from the estimation of Equation 6, where the sample is restricted to homicides committed against White, Black or Hispanic victims that are committed in Chicago between 2001 and 2023. The dependent variable is either a dummy for whether the victim i, who died at time y has been mentioned by local newspapers by the third day after the event (Columns 1 through 3) or the the total number of mentions by the third day (Columns 4 through 6). Columns 2 and 4 control for other demographics of the victim (age, gender), time variables that affect whether the news can appear in the printed paper and media-related time trends (day of the week, hour of day, month, year), or the newsworthiness of the homicide (weapon, whether it is gang related, whether is is domestic violence related). Columns 3 and 6 further add fixed effects for the tract where the incident occurred. Given the skew in the distribution of total mentions, a Poisson model is used for this second dependent variable rather than OLS. Effects in percentage terms with related standard errors are also reported for Models 4 through 6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

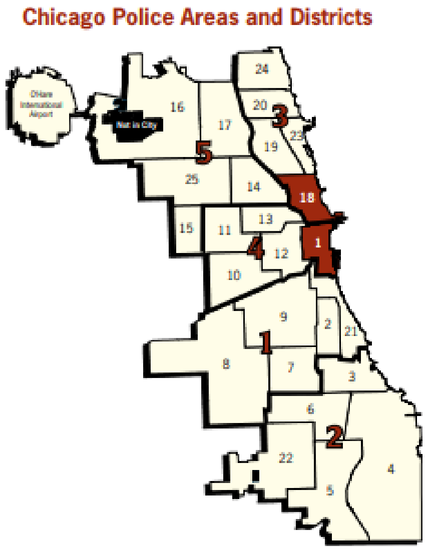
Appendix A: Supplemental Figures and Tables

Figure A.1: The Arrest-to-Court Pipeline in Cook County, IL

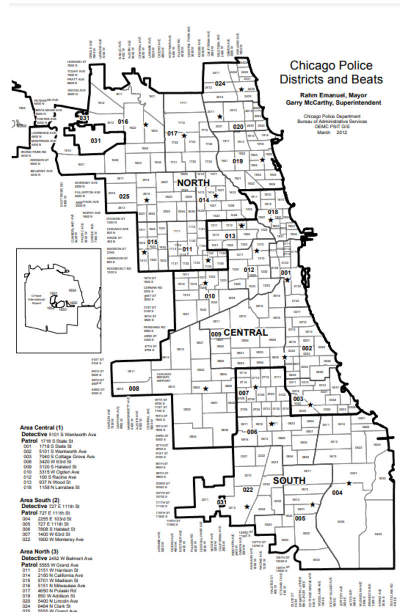


Notes: This figure shows the process that goes from entry to exit in the criminal justice system in Cook County, IL. Source: Cook County State Attorney's Office.

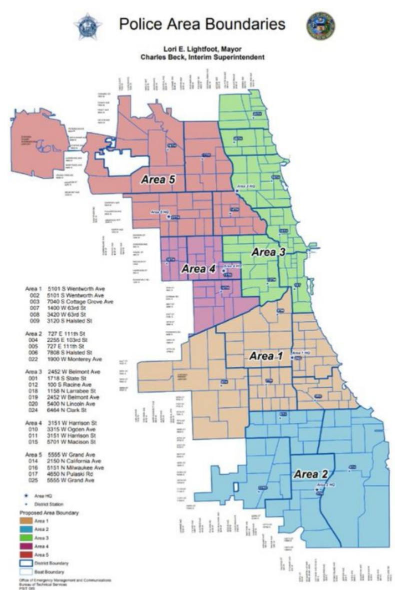
Figure A.2: Map of the City of Chicago with Detective Areas and Districts Labeled



(a) Before Closures in 3/2012



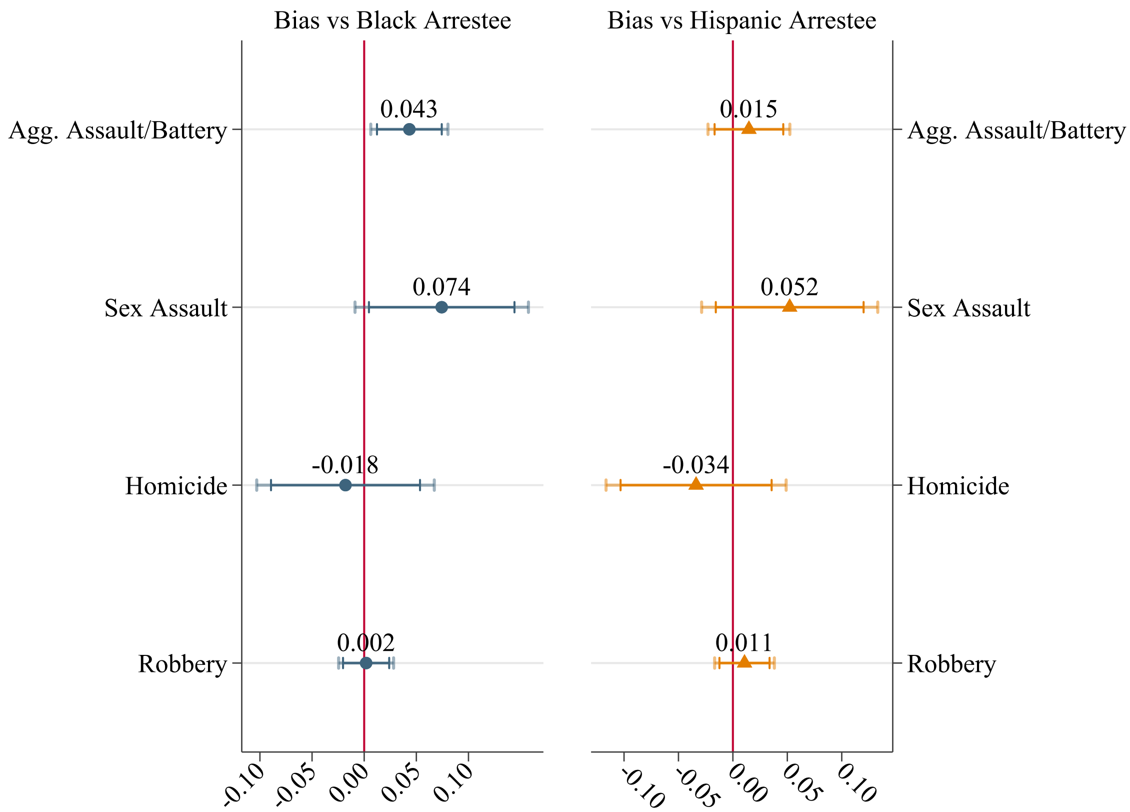
(b) After Closures in 3/2012 and Before Reopenings in 4/2020



(c) After Reopenings in 4/2020

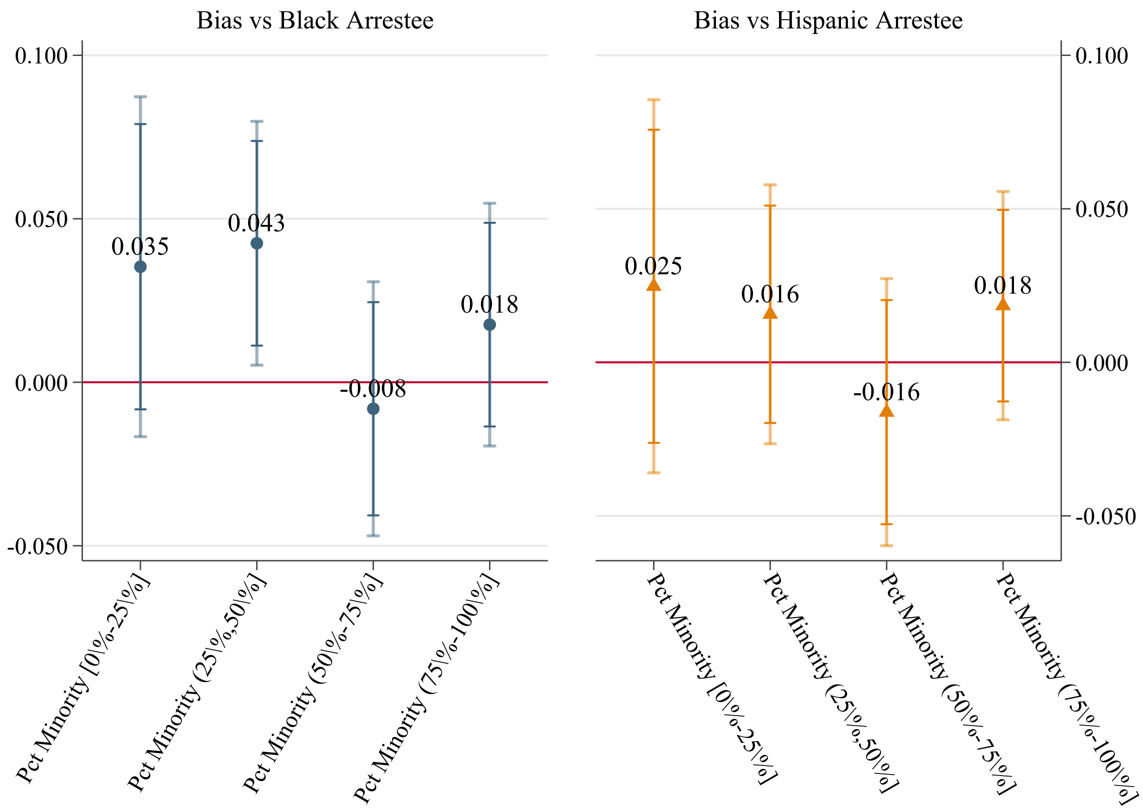
Notes: This figure shows the evolution of the detective areas' boundaries over time in Chicago, before closures in 3/2012 (Panel a), after closures in 3/2012 and before reopenings in 4/2020, and after reopenings in 2/2020

Figure A.3: Heterogeneity in Racial Bias by Type of Violent Crime



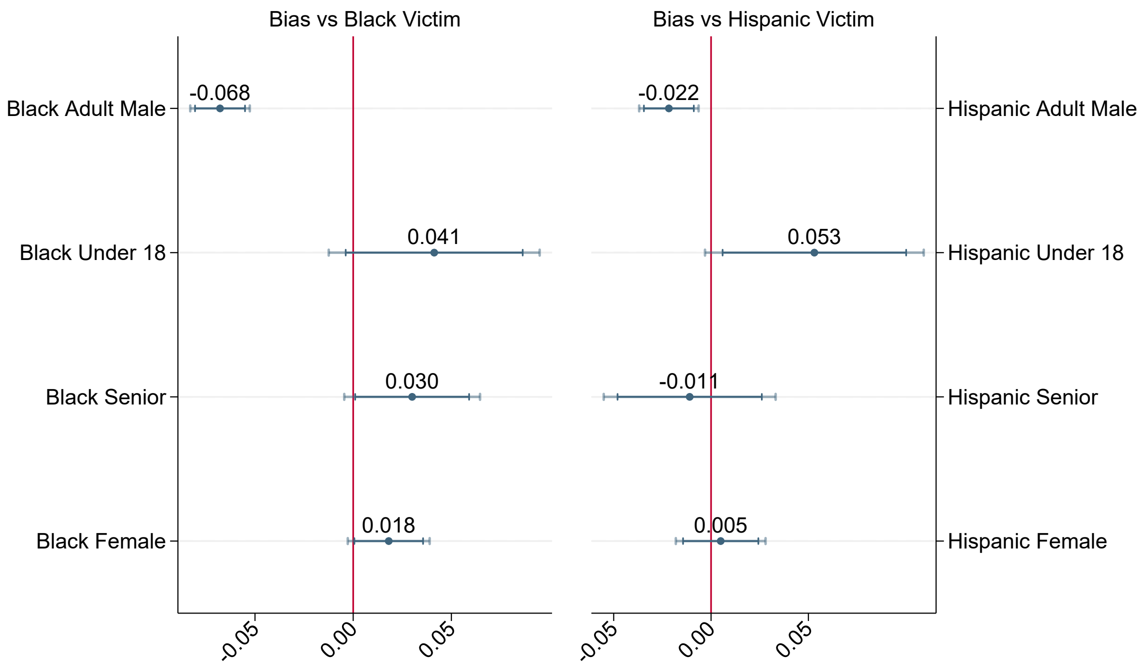
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5 by type of violent crime. Race dummies for victims and arrestees are interacted with fixed effects for the type of violent crime. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Figure A.4: Heterogeneity in Racial Bias by Racial Makeup of the Neighborhood



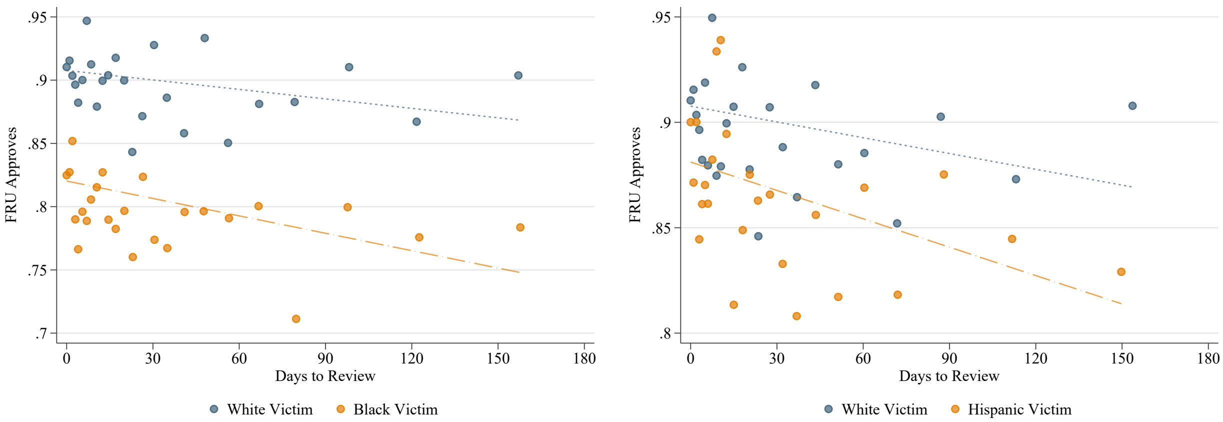
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5 by type of violent crime. Race dummies for victims and arrestees are interacted with fixed effects for the quartile of the share of minority residents in the Census tract. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Figure A.5: Heterogeneity in Racial Bias by Gender and Age of the Victim



Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5. Race dummies for victims are interacted with dummies for whether the victim is female, under 18 years old, or 60 years old or above. This figure shows the coefficients on Black and Hispanic victims, separately.

Figure A.6: Relationship between Acceptance and Days to Clearance

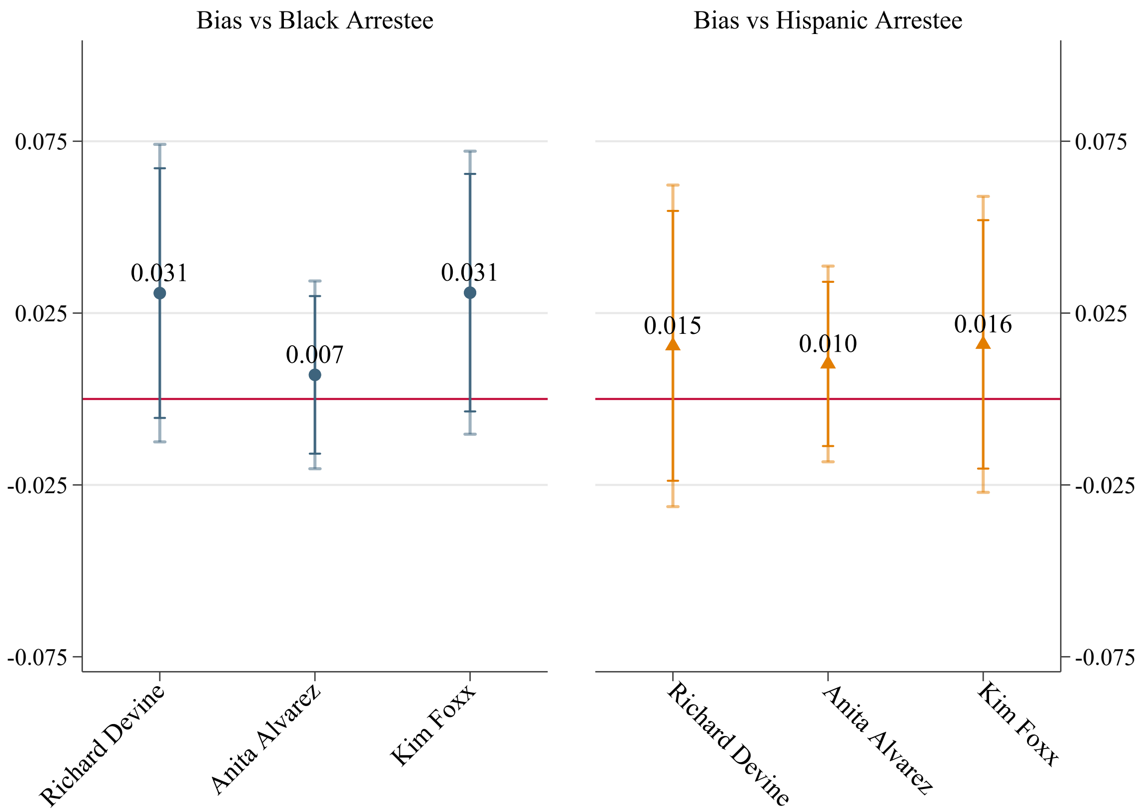


(a) White Victim vs. Black Victim

(b) White Victim vs. Hispanic Victim

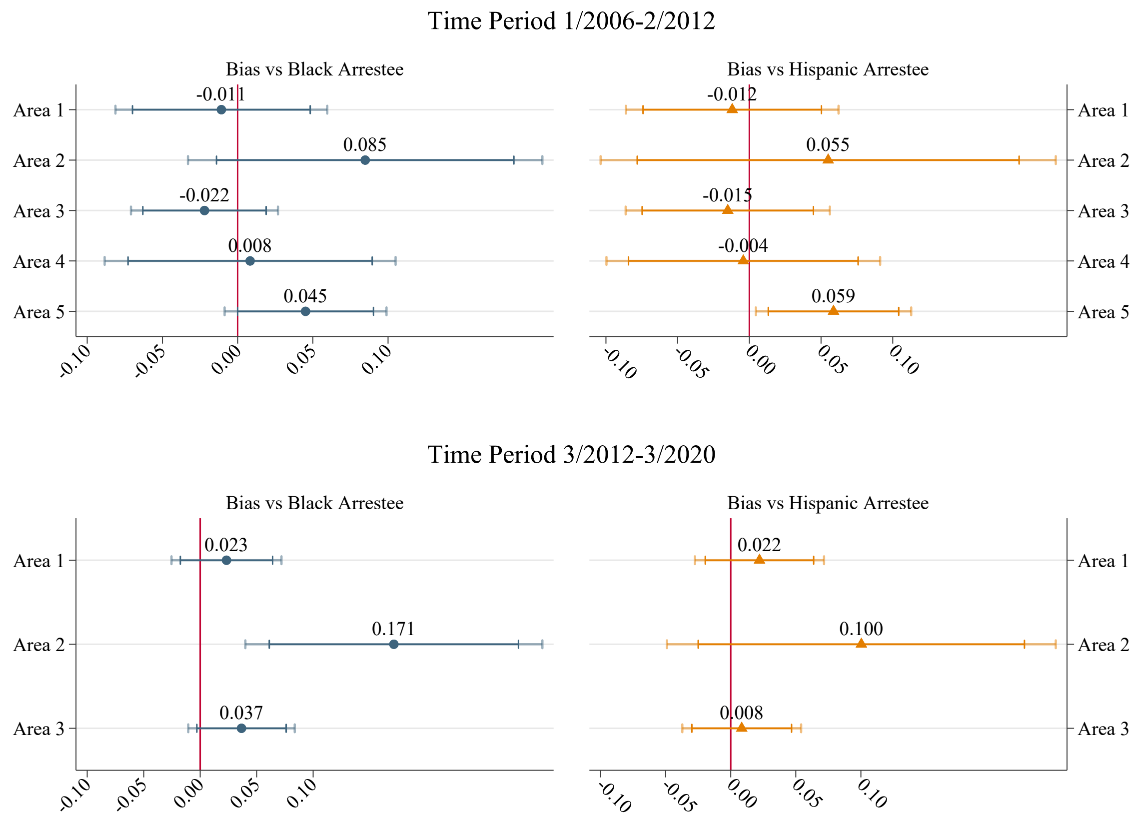
Notes: This figure plots a bin scatter of the relationship between acceptance by FRU and days elapsed from report of the crime to review, by race of the victim. The range of days to review is truncated at 180 days.

Figure A.7: Heterogeneity in Racial Bias by SA's Term



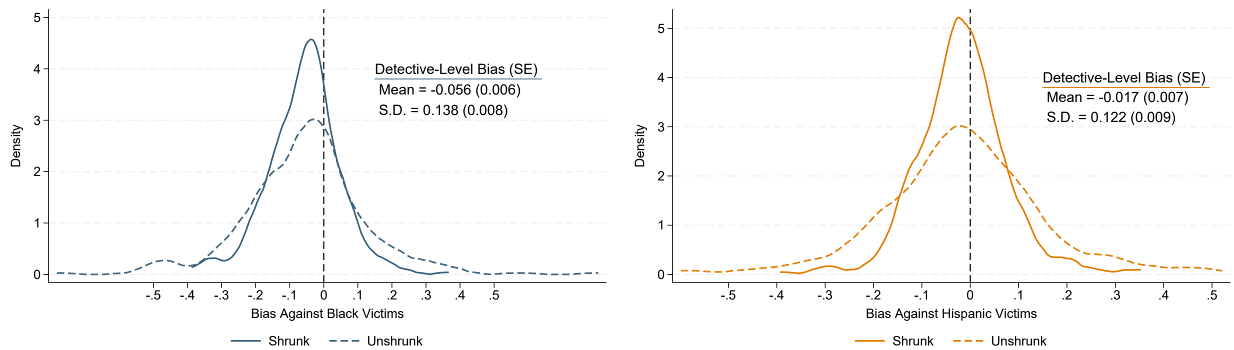
Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5 by SA leading the CCSAO. Race dummies for victims and arrestees are interacted with fixed effects for the SA leading the CCSAO. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Figure A.8: Heterogeneity in Racial Bias by Detective Area



Notes: This figure plots the coefficient from a heterogeneous version of the model used in Column 6 of Table 5. Race dummies for victims and arrestees are interacted with fixed effects for detective area where the crime occurred and investigated. This figure shows the coefficients on Black and Hispanic arrestees, separately.

Figure A.9: Detective-level Estimates of Bias

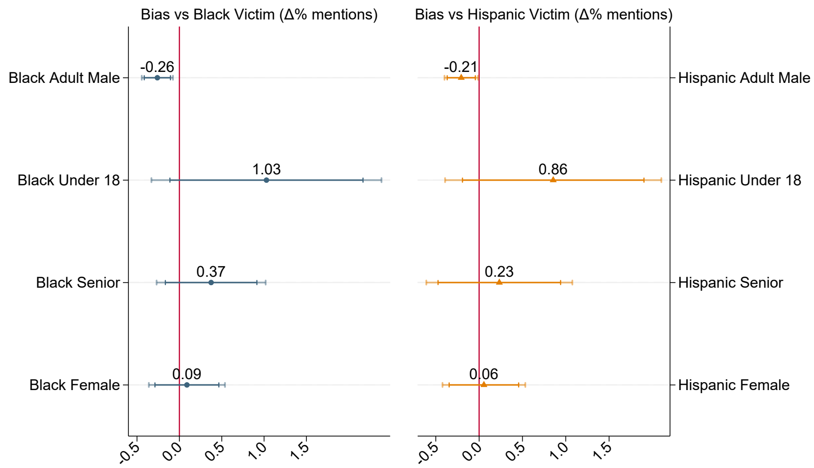


(a) White Victim vs. Black Victim

(b) White Victim vs. Hispanic Victim

Notes: This figure plots the distribution of detective-level bias, shrunk and unshrunk. Detective-level estimates of bias are obtained from Model 5, specifically as the interaction between dummies for the race of the victim with detective fixed effects. The hyper-parameters of the prior distribution (mean and standard deviation) and shrunken estimates are obtained following Walters (2024) and Morris (1983). See Appendix B.3.2 for details. Standard errors, reported in parentheses, are obtained by bootstrapping posterior estimates of detective-level bias obtained using Morris (1983)'s Empirical Bayes approach.

Figure A.10: Heterogeneity in Coverage Bias by Age and Gender of the Victim



Notes: This figure plots the coefficient from a heterogeneous version of Equation 6. The outcome is the number mentions in the first three days after the homicide. The coefficients are estimated using a Poisson model and then transformed to be percentage point differences. Race dummies for victims are interacted with dummies for whether the victim is female, under 18 years old, or 60 years old or above. This figure shows the coefficients on Black and Hispanic victims, separately.

Table A.1: Correlation Between Victim and Detective Characteristics

	(1) Black Det.	(2) Hispanic Det.	(3) White Det.	(4) Leave-Out Cl. Rate	(5) Leave-Out Arrest Cl. Rate	(6) Black Det.	(7) Hispanic Det.	(8) White Det.	(9) Leave-Out Cl. Rate	(10) Leave-Out Arrest Cl. Rate
Black Victim	0.052***	-0.010	-0.038***	0.004***	-0.002***	0.017***	-0.002	-0.016***	-0.002***	-0.001***
	(0.009)	(0.011)	(0.013)	(0.001)	(0.001)	(0.003)	(0.003)	(0.004)	(0.001)	(0.000)
Hispanic Victim	-0.005	0.035***	-0.030***	-0.003**	-0.000	0.003*	0.019***	-0.021***	-0.002***	-0.000
	(0.004)	(0.010)	(0.011)	(0.001)	(0.001)	(0.002)	(0.004)	(0.004)	(0.001)	(0.000)
Area × Time × Crime Type FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Case Controls						✓	✓	✓	✓	✓
Observations	619645	619645	619645	619645	619645	619645	619645	619645	619645	619645

Notes: This table shows the correlation between victim race and the characteristics of the detective first assigned to the case. The regressions condition on conditional on area-by-time-by-crime type fixed effects, which is the level of assignment, and Columns 6 to 10 add case controls as well. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. The dependent variables are the race of the detective, a leave-one-out measure of clearance rate and a leave-one-out measure of arrest clearance rate. Standard errors are clustered at the detective level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Disparities in Clearance Probability for crimes reported from 2006 onward

	(1) Cleared	(2) Cleared	(3) Cleared - No Victim Refusal	(4) Cleared - No Victim Refusal	(5) Cleared - Arrest and Prosecution	(6) Cleared - Arrest and Prosecution
Black Victim	-0.035***	-0.032***	-0.046***	-0.045***	-0.038***	-0.038***
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Hispanic Victim	-0.027***	-0.025***	-0.021***	-0.020***	-0.014***	-0.013***
	(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Pct Black (25%,50%]	0.000		-0.002		-0.002	
	(0.004)		(0.003)		(0.003)	
Pct Black (50%-75%]	-0.016***		-0.015***		-0.011***	
	(0.004)		(0.004)		(0.003)	
Pct Black (75%-100%]	-0.031***		-0.030***		-0.023***	
	(0.004)		(0.004)		(0.003)	
Pct Hispanic (25%,50%]	-0.007**		-0.007**		-0.003	
	(0.003)		(0.003)		(0.003)	
Pct Hispanic (50%-75%]	-0.022***		-0.018***		-0.012***	
	(0.004)		(0.004)		(0.003)	
Pct Hispanic (75%-100%]	-0.033***		-0.028***		-0.018***	
	(0.005)		(0.004)		(0.003)	
Ln(Median Per Capita Income)	0.003		0.006***		0.006***	
	(0.002)		(0.002)		(0.002)	
Area × Time × Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
Tract FE		✓		✓		✓
White Victim Mean	0.2788	0.2788	0.2193	0.2193	0.1644	0.1644
Observations	461,951	461,951	461,951	461,951	461,951	461,951
R^2	0.1817	0.1844	0.1205	0.1233	0.1052	0.1081

Notes: This table shows the results from the estimation of Model 1. The outcome in Columns 1 and 2 is overall clearance; the outcome in Columns 3 and 4 is an indicator for a clearance not achieved due to the victim's refusal; the outcome in Columns 3 and 4 is an indicator for a clearance achieved explicitly via arrest and prosecution. Case controls include whether the crime is gang-related, whether the crime is domestic violence-related, fixed effects for the weapon used in the crime, fixed effects for the premises where the crime occurred (e.g. street or residence), whether it was daylight or dark at the time of the crime, the day of the week when the crime occurred, the initial priority of the 911 call when the crime occurred, and how the crime was reported. Each observation is weighted by the inverse of the total number of victims involved in the incident. Standard errors are clustered at the detective area-time-crime type level. The sample is restricted to crime reported from 2006 afterwards. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Characteristics of the Sample for the Outcome Test

	Homicide			***Non-Fatal Violent Crime***		
	White Victims	***Black Victims***	***Hispanic Victims***	***White Victims***	***Black Victims***	***Hispanic Victims***
Female Victim	0.266	0.164	0.159	0.400	0.431	0.412
Victim's Age	40.241	30.934	27.334	35.967	31.379	29.044
Black Suspect	0.415	0.968	0.210	0.661	0.961	0.404
Hispanic Suspect	0.297	0.031	0.718	0.172	0.029	0.546
White Suspect	0.317	0.004	0.096	0.178	0.013	0.063
Female Suspect	0.076	0.100	0.057	0.074	0.097	0.066
Assault				0.030	0.078	0.059
Battery				0.214	0.331	0.213
Robbery				0.659	0.443	0.549
Sexual Assault				0.097	0.148	0.179
FRU Approves	0.874	0.812	0.835	0.906	0.808	0.872
Total Observations	239	2,497	600	6,794	17,574	8,988

Notes: This table shows the average characteristics of the victims of violent crime, with related information about the incident, arrestee, investigation and assigned detective, for those observations that enter the sample of the outcome test. The information is shown separately for White, Black and Hispanic victims and for homicides and nonfatal violent crime. The sample includes offenses reported between 2001 and 2023 and arrestees taken into custody from 2006 and 2023 and charged by the police for at least one felony. Only White, Black and Hispanic victims, and White, Black and Hispanic arrestees are kept in the sample.

Table A.4: Top Charge Guilty

	(1) Top Charge Guilty	(2) Top Charge Guilty	(3) Top Charge Guilty	(4) Top Charge Guilty
Black Victim	-0.116***	-0.086***	-0.081***	-0.081***
	(0.010)	(0.012)	(0.012)	(0.012)
Hispanic Victim	-0.058***	-0.032***	-0.024**	-0.022*
	(0.010)	(0.011)	(0.011)	(0.012)
Number of Charges	0.021***	0.023***	0.023***	0.023***
	(0.002)	(0.003)	(0.002)	(0.002)
Pct Black (25%,50%]		-0.007		
		(0.016)		
Pct Black (50%-75%]		-0.030		
		(0.020)		
Pct Black (75%-100%]		-0.033*		
		(0.020)		
Pct Hispanic (25%,50%]		-0.016		
		(0.016)		
Pct Hispanic (50%-75%]		-0.022		
		(0.019)		
Pct Hispanic (75%-100%]		-0.031		
		(0.022)		
Ln(Median Per Capita Income)		0.017		
		(0.011)		
Black Arrestee				0.040**
				(0.018)
Hispanic Arrestee				0.016
				(0.019)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.6339	0.6339	0.6339	0.6339
Observations	30,899	30,881	30,894	30,894
R^2	0.2768	0.2863	0.3185	0.3220

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the top charge brought against the arrestee is associated with a guilty ruling. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Appears at Arraignment

	(1) Arraignment	(2) Arraignment	(3) Arraignment	(4) Arraignment
Black Victim	-0.077***	-0.069***	-0.068***	-0.067***
	(0.009)	(0.010)	(0.010)	(0.011)
Hispanic Victim	-0.027***	-0.019**	-0.023**	-0.023**
	(0.009)	(0.009)	(0.010)	(0.010)
Number of Charges	0.021***	0.020***	0.020***	0.020***
	(0.002)	(0.002)	(0.002)	(0.002)
Pct Black (25%,50%]		-0.010		
		(0.013)		
Pct Black (50%-75%]		-0.003		
		(0.017)		
Pct Black (75%-100%]		-0.023		
		(0.017)		
Pct Hispanic (25%,50%]		0.003		
		(0.013)		
Pct Hispanic (50%-75%]		-0.002		
		(0.015)		
Pct Hispanic (75%-100%]		-0.022		
		(0.018)		
Ln(Median Per Capita Income)		0.008		
		(0.010)		
Black Arrestee				0.009
				(0.014)
Hispanic Arrestee				0.008
				(0.015)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age		✓	✓	✓
Arrestee Gender and Age		✓	✓	✓
Tract FE			✓	✓
Difficulty Controls		✓	✓	✓
White Victim Mean	0.8297	0.8297	0.8297	0.8297
Observations	30,899	30,881	30,894	30,894
R^2	0.2623	0.2728	0.3027	0.3051

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the arrestee appears at arraignment. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Recidivism

	(1) 3 Years Violent Recidivism	(2) 3 Years Violent and Felony Recidivism	(3) 1 Year Violent Recidivism	(4) 1 Year Violent and Felony Recidivism
Black Victim	0.049***	0.037***	0.029***	0.020**
	(0.010)	(0.010)	(0.009)	(0.008)
Hispanic Victim	0.029***	0.025**	0.008	0.007
	(0.010)	(0.010)	(0.008)	(0.008)
Black Arrestee	0.057***	0.059***	0.027**	0.030**
	(0.015)	(0.014)	(0.013)	(0.012)
Hispanic Arrestee	0.016	0.017	0.010	0.009
	(0.015)	(0.014)	(0.012)	(0.012)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
White Victim Mean	0.2912	0.2735	0.1569	0.1431
Observations	39,834	39,834	39,834	39,834
R^2	0.2528	0.2429	0.2190	0.2130

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the arrestee is take in into custody for another index violent crime in the future in Columns 1 and 3, and for index violent crime that also involved a felony in Columns 2 and 4. Columns 1 and 2 consider a three year horizon, while Columns 3 and 4 consider a one year horizon. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Status of the Investigation

	Full Sample		1 Arrestee Involved	
	(1) FRU Approves	(2) Cleared - Arrest and Prosecution	(3) FRU Approves	(4) Cleared - Arrest and Prosecution
Black Victim	-0.058***	-0.022***	-0.061***	-0.024***
	(0.007)	(0.007)	(0.008)	(0.009)
Hispanic Victim	-0.016**	-0.006	-0.008	-0.001
	(0.007)	(0.007)	(0.008)	(0.009)
Black Arrestee	0.019*	-0.006	0.035***	0.000
	(0.011)	(0.011)	(0.013)	(0.013)
Hispanic Arrestee	0.012	0.012	0.015	0.015
	(0.011)	(0.012)	(0.013)	(0.014)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓
White Victim Mean	0.9103	0.8722	0.9103	0.8722
Observations	43,755	43,755	29,194	29,194
R^2	0.2968	0.2883	0.3390	0.3202

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the status of the investigation is clearance by arrest and prosecution. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Additional Robustness Checks

	1 Victim	1 Arrestee	1 Victim and 1 Arrestee	No Weighting	Detective Clustering	First Detective
	(1)	(2)	(3)	(4)	(5)	(6)
	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves	FRU Approves
Black Victim	-0.058***	-0.061***	-0.065***	-0.055***	-0.058***	-0.059***
	(0.008)	(0.008)	(0.010)	(0.006)	(0.007)	(0.007)
Hispanic Victim	-0.019**	-0.008	-0.013	-0.019***	-0.016**	-0.016**
	(0.008)	(0.008)	(0.010)	(0.006)	(0.006)	(0.007)
Black Arrestee	0.021*	0.035***	0.042***	0.008	0.019*	0.021*
	(0.012)	(0.013)	(0.015)	(0.010)	(0.010)	(0.011)
Hispanic Arrestee	0.013	0.015	0.018	0.009	0.012	0.014
	(0.013)	(0.013)	(0.015)	(0.010)	(0.011)	(0.011)
Area × Review Month-Year × Crime Type FE	✓	✓	✓	✓	✓	✓
Detective FE	✓	✓	✓	✓	✓	✓
Victim Gender and Age	✓	✓	✓	✓	✓	✓
Arrestee Gender and Age	✓	✓	✓	✓	✓	✓
Tract FE	✓	✓	✓	✓	✓	✓
Difficulty Controls	✓	✓	✓	✓	✓	✓
White Victim Mean	0.9103	0.9103	0.9103	0.9103	0.9103	0.9113
Observations	29,528	29,194	20,776	43,755	43,755	43,857
R^2	0.3297	0.3390	0.3709	0.2943	0.2968	0.2928

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the status of the investigation is clearance by arrest and prosecution. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: Estimates in the Sample used to Estimate Detective Level Bis

	(1) FRU Approves	(2) FRU Approves
Black Victim	-0.067***	-0.058***
	(0.008)	(0.009)
Hispanic Victim	-0.018**	-0.016**
	(0.007)	(0.008)
Black Arrestee		0.014
		(0.013)
Hispanic Arrestee		0.015
		(0.014)
Area × Review Month-Year × Crime Type FE	✓	✓
Detective FE	✓	✓
Victim Gender and Age		✓
Arrestee Gender and Age		✓
Tract FE		✓
Difficulty Controls		✓
White Victim Mean	0.9103	0.9103
Observations	22,053	22,043
R^2	0.2430	0.3072

Notes: This table shows the results from the estimation of Equation 4 in the sample to estimate detective level bias as displayed in Figure A.9. Column 1 uses the same model used in Column 4 of Table 4. Column 2 uses the same model used in Column 6 of Table 5. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. The main regressors of interest are dummies that take value 1 if the victim is Black or Hispanic. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.10: Heterogeneity by Detective's Race

	(1) FRU Approves	(2) FRU Approves
Black Victim	-0.059***	-0.059***
	(0.008)	(0.007)
Hispanic Victim	-0.018**	-0.018**
	(0.008)	(0.007)
Victim and Detective Race Match	-0.001	
	(0.007)	
Black Victim × Black Detective		0.006
		(0.018)
Hispanic Victim × Hispanic Detective		0.001
		(0.014)
Area × Review Month-Year × Crime Type FE	✓	✓
Detective FE	✓	✓
Tract FE	✓	✓
Difficulty Controls	✓	✓
Victim Gender and Age	✓	✓
Defendant Race, Gender and Age	✓	✓
Observations	42,746	42,746
R^2	0.2976	0.2976

Notes: This table shows the results from the estimation of Equation 4 with additional controls. The dependent variable is a dummy that takes value 1 if the case is accepted by the FRU. Column 1 's main regressor is a dummy that takes value 1 if the race of the detective and the race of the victim match. Column 2 disaggregates this further by creating separate interactions for Black and Hispanic detective-victim pairs. Observations are weighted by the inverse of the number of victims and arrestees involved in the incident. Standard errors are clustered at the area-time-crime type level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix B: Theory Appendix

B.1 Proof of Proposition 1

We can show that if there is a gap in success rates between non-White and White cases that are solved and handed over to the prosecutor after the same number of days t_i^*, the investigator displays racial bias:

$$\begin{aligned}
&E[Y_i \mid RV_i = m, D_i = 1] - E[Y_i \mid RV_i = w, D_i = 1] \\
=&E[Y_i^*(t) \mid RV_i = m, T_i = t_i^*] - E[Y_i^*(t) \mid RV_i = w, T_i = t_i^*] \\
=&E[Y_i^*(t) \mid RV_i = m, p_i(t) = c(m)] - E[Y_i^*(t) \mid RV_i = w, p_i(t) = c(w)] \\
=&E[E[Y_i^*(t) \mid RV_i, U_i] \mid RV_i = m, p_i(t) = c(m)] - \\
&E[E[Y_i^*(t) \mid RV_i, U_i] \mid RV_i = w, p_i(t) = c(w)] \\
=&E[p_i(t) \mid RV_i = m, p_i(t) = c(m)] - E[p_i(t) \mid RV_i = w, p_i(t) = c(w)] \\
=&c(m) - c(w).
\end{aligned} \tag{7}$$

The first equality follows because $Y_i = Y_i^*(T_i)$ and $T_i = t_i^* \iff D_i = 1$. The second equality follows because $T_i = t_i^* \iff p_i(t_i^*) = c(rv)$. The third equality follows by the law of iterated expectations. The fourth equality follows because $p_i(t) = E[Y_i^*(t)|RV_i, U_i]$. The fifth equality follows by the conditioning on $p_i(t)$.

B.2 Microfoundation of the thresholds $c(RV_i)$

A detective decides whether to bring the case of victim i to the DA's office, represented by $D_i \in \{0, 1\}$. If the case is submitted, this can either be successful if the DA accepts the charges against the suspect related to the case of victim i ($Y_i^* = 1$) or a failure if the DA rejects charges ($Y_i^* = 0$). In each period, the detective has beliefs $p_i(t)$ over the success of the case. Upon clearance, she receives utility κ regardless of the success status of the case. She dislikes making both false positives ($D_i = 1, Y_i^* = 0$) and false negatives ($D_i = 0, Y_i^* = 1$). These costs can be race-dependent and can be defined as $\beta(RV_i)$ and $\gamma(RV_i)$, respectively, where $RV_i \in \{w, m\}$ denotes the race of the victim. Her utility in period t is therefore:

$$U(t) = \kappa - \beta(RV_i)D_i(t)(1 - Y_i^*(t)) - \gamma(RV_i)(1 - D_i(t))Y_i^*(t), \tag{8}$$

with expected utility for different actions $d \in \{0, 1\}$ being

$$\begin{aligned}
E[U(t)|D_i(t) = 0] &= \kappa - \gamma(RV_i)p_i(t) \\
E[U(t)|D_i(t) = 1] &= \kappa - \beta(RV_i)(1 - p_i(t))
\end{aligned} \tag{9}$$

Her optimal decision rule is then:

$$
\begin{aligned}
D_i &= \mathbf{1}[-\beta(1-p_i(t)) \geq -\gamma p_i(t)] \\
&= \mathbf{1}[p_i(t) \geq \frac{\beta(RV_i)}{\beta(RV_i)+\gamma(RV_i)} = c(RV_i)]
\end{aligned}
\tag{10}
$$

From the last equation, we see that $c(RV_i)$ can be recast as the relative cost of type I error upon submission at period t of a case of victim of race RV_i. We can therefore redefine racial bias, $c(m) < c(w)$, as the detective having a lower relative cost of an unsuccessful submission when the victim is Black rather than White.

This difference in relative costs can arise in several ways. The detective can perceive, for example, race-neutral costs of false negatives ($\gamma(m) = \gamma(w)$) but a lower cost of false positives for Black victims than for White victims ($\beta(m) < \beta(w)$): submitting a shoddy investigation at time t is less costly if the victim is Black rather than White. Alternatively, the detective can perceive, for example, race-neutral costs of false negatives ($\beta(m) = \beta(w)$) but a larger cost of false positives for Black victims than for White victims ($\gamma(m) > \gamma(w)$): the detective dislikes keeping potentially successful cases with Black victims open more than those of White victims since keeping a case open entails additional investigative effort, which under discrimination against Black victims is more costly for the latter group than for White victims. Moreover, by devoting additional time to one case, the detective expends a scarce resource that could be allocated to other investigations. The detective may perceive the cost of a type II error to be higher if she regards spending time on a case involving a Black victim as a particularly inefficient use of resources. A combination of these two scenarios ($\beta(m) < \beta(w)$ and $\gamma(m) > \gamma(w)$) can also yield a lower relative cost of failure for cases with Black victims than for cases with White victims.

B.3 A unified framework for racial bias against minority victims and minority suspect

B.3.1 Discrimination against minority suspects

The model used to derive the main predictions for racial bias against minority victims can be adapted to back out predictions regarding minority suspects as well. To do so, I momentarily ignore the presence of victims. The detective decides again when to submit the case of suspect k, of race $RS_k \in \{w, m\}$, in a way to minimize type I and II errors. This means that the decision rule followed by the detective is mathematically the same:

$$
D_k(t) = \mathbf{1}[p_k(t) \geq \frac{\beta(RS_k)}{\beta(RS_k)+\gamma(RS_k)} = c(RS_k)] \tag{11}
$$

The main difference is in the nature of type I and type II errors, which also yield a different prediction for racial bias. Submitting a case that is rejected in this case means that a suspect goes free. Assuming the detective pursues suspect she believes are guilty, this corresponds to letting a guilty person go. In addition, a type I error also includes public-safety concern since the suspect might commit a new crime, which could be perceived as especially likely if the detective believed the individual was guilty in the first place. If a detective perceives minority suspects to be particularly deserving of justice and imprisonment and/or more dangerous if allowed to be free, the cost of a false positive is higher among them, i.e. $\beta(m) > \beta(w)$.

On the other hand, delaying a submission and spending extra time on a case entails a resource and opportunity cost: the detective could spend that time investigating other cases. If the detective finds working on crimes linked to a minority suspects relatively more worth than expending effort on cases involving White suspects, a false negative is less costly for the former, i.e. $\gamma(m) < \gamma(w)$. A potential concern for public safety might arise also when considering spending additional time on case: an individual who is considered dangerous is spending an extra day in the community and might commit a new crime. However, because a rejection upon submission implies freedom for a much longer horizon in expected terms than a delay in submission, I view recidivism mostly as a type I concern.

Overall, the relationships between costs of type I and II errors described above imply that a biased detective desires a higher standard of quality at submission for minority suspects than for White suspects, i.e. $c(m) > c(w)$. This requires making two simplifying assumptions: the detective assumes that the suspect is guilty - which also abstracts away from potentially wrongful convictions, and recidivism is mainly a type I concern. Empirically, this means observing positive coefficients on dummies for Black and Hispanic suspects.

B.3.2 Combining victims and suspects

The detective must ultimately choose a single stopping point that balances her preferences toward both victims and suspects. This overall decision rule can be derived as a convex combination of the victim-specific and suspect-specific thresholds outlined in the previous sections.

Define $\beta_v(RV_i)$ and $\gamma_v(RV_i)$ as the victim-related costs of type I and type II errors, respectively, and $\beta_s(RS_k)$ and $\gamma_s(RS_k)$ as the corresponding suspect-related costs. Each is measured in the same unit of social or institutional cost (e.g., expected harm, wasted resources, reputational cost). Assume that the overall cost of each type of error is a weighted average of a victim-related and suspect-related cost of the same type of error. The detective places weights λ_v and $1 - \lambda_v$ on each of the two, respectively:

$$\beta(RV_i, RS_k) \equiv \lambda_v\, \beta_v(RV_i) + (1-\lambda_v)\, \beta_s(RS_k),$$
$$\gamma(RV_i, RS_k) \equiv \lambda_v\, \gamma_v(RV_i) + (1-\lambda_v)\, \gamma_s(RS_k). \tag{12}$$

The detective's utility in period t is then:

$$U(t) = \kappa - \beta(RV_i, RS_k)\, D_{ik}(t)\big(1 - Y^*_{ik}(t)\big) - \gamma(RV_i, RS_k)\,\big(1 - D_{ik}(t)\big)Y^*_{ik}(t). \tag{13}$$

This yields the usual decision rule

$$D_{ik}(t) = \mathbf{1}\left[p_{ik}(t) \geq \frac{\beta(RV_i, RS_k)}{\beta(RV_i, RS_k) + \gamma(RV_i, RS_k)} = c(RV_i, RS_k)\right] \tag{14}$$

To simplify notation, let $S_v(RV_i) \equiv \beta_v(RV_i) + \gamma_v(RV_i)$ and $S_s(RS_k) \equiv \beta_s(RS_k) + \gamma_s(RS_k)$ denote the total stakes for each party.

The overall threshold can be written as a convex combination of these party-specific thresholds:

$$c(RV_i, RS_k) = \lambda^*_v(RV_i, RS_k)\, c_v(RV_i) + \big(1 - \lambda^*_v(RV_i, RS_k)\big)\, c_s(RS_k), \tag{15}$$

with the *effective* weight

$$\lambda^*_v(RV_i, RS_k) \equiv \frac{\lambda_v\, S_v(RV_i)}{\lambda_v\, S_v(RV_i) + (1-\lambda_v)\, S_s(RS_k)} \tag{16}$$

which is itself a function of victim and suspect race. For ease of exposition, we can normalize the victim's and suspect's stakes to be equal, so that $\lambda^*_v(RV_i, RS_k) = \lambda_v$.

Equation (15) shows that the overall stopping point is a convex combination of the victim-specific and suspect-specific thresholds. The same victim can face different outcomes depending on the identity of the suspect ($c(rv, rs) \neq c(rv, rs')$), and similarly for suspects of a given race ($c(rv, rs) \neq c(rv', rs)$). Combining the definitions of bias against minority victims and minority suspects from the previous sections, we can expect the following ranking across victim–arrestee dyads:

$$c(w, m) > c(w, w) \lesseqgtr c(m, m) > c(m, w), \tag{17}$$

where minority-on-White crime is the most harshly sanctioned, and White-on-minority crime is the least likely to be accepted. The relationship between the thresholds picked for White-on-White crime and minority-on-minority is instead ambiguous.

Appendix C: Empirical Bayes

I estimate racial bias against Black and Hispanic victims for each individual detective by running the following regression:

$$Y_{ik} = \alpha + \beta_j^B \, BlackVictim_i + \beta_j^H \, HispanicVictim_i + \gamma' X_{ik} + \delta_j + \delta_{ayc} + \varepsilon_{ik}, \tag{18}$$

I denote the OLS estimates from this regression by $\hat{\beta}_j^B$ and $\hat{\beta}_j^H$. I then adopt an Empirical Bayes procedure to obtain a bias-corrected estimate of the true variation in bias.

I assume that the detective-level bias estimates as noisy measurements of latent "true" biases:

$$\hat{\beta}_j^r \mid \beta_j^r \sim N(\beta_j^r, s_j^r), \qquad \beta_j^r \sim N(\mu_r, \sigma_r^2), \qquad r \in \{\text{Black, Hispanic}\}, \tag{19}$$

with s_j^r replaced by its estimate $\hat{s}_j^r$. I estimate the hyperparameters μ_r and σ_r^2 following Walters (2024):

$$\hat{\mu}_r = \frac{1}{J}\sum_{j=1}^{J} \hat{\beta}_j^r, \qquad \hat{\sigma}_r^2 = \text{Var}(\hat{\beta}_j^r) - \frac{1}{J}\sum_{j=1}^{J} \hat{s}_j^r, \tag{20}$$

where J is the number of detectives.

I estimate the sampling variances of each $\hat{\theta}_j^r$, s_j^r, using the residuals from equation 18, $\hat{\varepsilon}_{ik}$. Specifically, for each detective j and each race $r \in \{\text{Black, Hispanic, White}\}$, I compute the residual variance within that detective-race cell,

$$\hat{\sigma}_{j,r}^2 = \frac{1}{N_{j,r} - 1} \sum_{\substack{j=1, \\ VR_i = r}}^{J} D_{j(ik)} \hat{\varepsilon}_{ik}^2, \tag{21}$$

where $N_{j,r}$ is the number of observations for detective j and race r, and $D_{j(i,k)}$ is an indicator that takes value one if the case of the victim-arrestee pair (i, k) was handled by detective j. I allow the variance to differ across detectives and across races. Under these assumptions, the variance of the detective-level Black–White gap is well approximated by the variance of a difference in sample means. This means that the sampling variances $\hat{s}_j^r$ as follows:

$$\hat{s}_j^B = \frac{\hat{\sigma}_{j,B}^2}{N_{j,B}} + \frac{\hat{\sigma}_{j,W}^2}{N_{j,W}} \qquad \hat{s}_j^H = \frac{\hat{\sigma}_{j,H}^2}{N_{j,H}} + \frac{\hat{\sigma}_{j,W}^2}{N_{j,W}}. \tag{22}$$

With estimates of the true between-detective variance in bias ($\hat{\sigma}_r^2$) and the sampling error of each individual estimate ($\hat{s}_j^r$), I can form a posterior estimate of detective-level bias by shrinking the raw coefficients towards the common mean proportionally to their relative noise. The resulting empirical Bayes estimator of detective j's bias against group r is the posterior mean,

$$\tilde{\beta}_j^r = \lambda_j^r \hat{\beta}_j^r + (1 - \lambda_j^r)\hat{\mu}_r, \qquad \lambda_j^r = \frac{\hat{\sigma}_r^2}{\hat{\sigma}_r^2 + \hat{s}_j^r}. \tag{23}$$